

Space–Time Algebra

A compact development of paravectors and their physical applications

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Abstract

This document develops a three-generator sigma algebra and its complex paravectors, then applies them to relativity, electromagnetism, and quantum mechanics. Natural units $\hbar = c = 1$ and signature $(+, -, -, -)$ are used.

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1 The algebra at a glance

The generators σ_n are algebraic objects with ordered multiplication rules. They are not ordinary vector components: their products generally do not commute, so the order of the factors is part of every expression. The following panel previews the generators, their product, and the general complex element developed below.

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} &= \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 &= -1, \\ \boldsymbol{\sigma} &= \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z &= (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{aligned}$$

Ordinary three-vector operations use

$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3, \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \quad (1.2)$$

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (1.3)$$

Thus the square of a vector always denotes its self dot product.

The set $\{1, \boldsymbol{\sigma}, \mathbf{i}\boldsymbol{\sigma}, \mathbf{i}\}$ is a complete basis, displayed by type as

$$\underbrace{1}_{\text{scalar}}, \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}}, \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}}, \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

Anticommutation gives $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$. The pseudoscalar is central: moving σ_n through the other two generators produces two sign changes, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}. \quad (1.4)$$

Thus its order of multiplication does not matter.

The commutator is

$$[X, Y] := XY - YX \quad (1.5)$$

$$[Y, X] = -[X, Y]. \quad (1.6)$$

Thus $[X, Y] = 0$ exactly when X and Y commute. For $n, m, k \in \{1, 2, 3\}$, define the Kronecker delta by

$$\delta_{nm} := \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \quad (1.7)$$

and the Levi-Civita symbol by

$$\epsilon_{nmk} := \begin{cases} +1, & (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two or more indices are equal.} \end{cases} \quad (1.8)$$

The ordered sigma multiplication rule is then For $n, m \in \{1, 2, 3\}$,

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9)$$

$$[\sigma_n, \sigma_m] = 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k. \quad (1.10)$$

2 Hermitian and sigma-conjugate operators

Define the two operations by

$$(\cdot)^{\mathbf{H}} := \text{reverse the order of every product of sigma generators}, \quad (2.1)$$

$$(\cdot)^* := \text{replace every } \sigma_n \text{ by } -\sigma_n \text{ without changing product order}. \quad (2.2)$$

Thus, for two generators,

$$(\sigma_m \sigma_n)^{\mathbf{H}} = \sigma_n \sigma_m, \quad (2.3)$$

$$(\sigma_m \sigma_n)^* = (-\sigma_m)(-\sigma_n). \quad (2.4)$$

Both operations act term by term on sums. Their actions on the four basis types are

type representative	scalar 1	vector σ_n	bivector $\mathbf{i} \sigma_n$	pseudoscalar $\mathbf{i}$
$(\cdot)^*$	+	−	+	−
$(\cdot)^{\mathbf{H}}$	+	+	−	−
$(\cdot)^{* \mathbf{H}}$	+	−	−	+

In particular,

$$\sigma_n^* = -\sigma_n, \quad (2.5)$$

$$\sigma_n^{\mathbf{H}} = \sigma_n, \quad (2.6)$$

$$\mathbf{i}^* = -\mathbf{i}, \quad (2.7)$$

$$\mathbf{i}^{\mathbf{H}} = -\mathbf{i}. \quad (2.8)$$

For the bivectors, these actions combine to give

$$(\mathbf{i} \sigma_n)^* = \mathbf{i}^* \sigma_n^* = \mathbf{i} \sigma_n \quad (2.9)$$

$$(\mathbf{i} \sigma_n)^{\mathbf{H}} = \sigma_n^{\mathbf{H}} \mathbf{i}^{\mathbf{H}} = -\mathbf{i} \sigma_n. \quad (2.10)$$

Thus sigma conjugation preserves the original order of the factors, whereas the Hermitian operator changes their order. For arbitrary algebra elements Z, Z' , these definitions give

$$(ZZ')^{\mathbf{H}} = Z'^{\mathbf{H}} Z^{\mathbf{H}}, \quad (ZZ')^* = Z^* Z'^*, \quad Z^{* \mathbf{H}} = Z^{\mathbf{H} *}.$$

Appendix A reproduces this as Hermitian conjugation in the Pauli-matrix representation.

3 Paravectors

3.1 Core operations

Let $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. First form the two real paravectors

$$X := a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.1)$$

$$Y := b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.2)$$

Their complex combination is the general complex paravector

$$Z := X + iY = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}. \quad (3.3)$$

Equivalently, collect its complex scalar and vector coefficients as

$$S := a + bi \quad (3.4)$$

$$\mathbf{V} := \mathbf{A} + \mathbf{B}i \quad (3.5)$$

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.6)$$

The corresponding coefficient-extraction operations are

$$\text{sc}(Z) := S \quad (3.7)$$

$$\text{vec}(Z) := \mathbf{V}. \quad (3.8)$$

Sigma conjugation changes the sign of every sigma generator without changing multiplied order; the Hermitian operator changes multiplied sigma order. For the complex coefficients, write

$$S^{\text{H}} = S^* := a - bi \quad (3.9)$$

$$\mathbf{V}^{\text{H}} = \mathbf{V}^* := \mathbf{A} - \mathbf{B}i. \quad (3.10)$$

For a paravector, the two operations are then

$$Z^* := S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (-\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (3.11)$$

$$Z^{\text{H}} := S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - bi) + (\mathbf{A} - \mathbf{B}i) \cdot \boldsymbol{\sigma}. \quad (3.12)$$

Their composition is the adjugate:

$$\text{adj}(Z) := Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.13)$$

Define the paravector trace and determinant by

$$\text{tr}(Z) := Z + \text{adj}(Z) = 2S, \quad (3.14)$$

$$\det(Z) := \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (3.15)$$

For any paravectors Z and Z' , the adjugate reverses product order:

$$\text{adj}(ZZ') = \text{adj}(Z') \text{adj}(Z). \quad (3.16)$$

Consequently, for $z^2 \neq 1$ where the inverse occurs,

$$\det(ZZ') = \text{adj}(ZZ')ZZ' \quad (3.17)$$

$$= \text{adj}(Z')(\text{adj}(Z) Z)Z' \quad (3.18)$$

$$= \det(Z) \det(Z'). \quad (3.19)$$

For any complex scalar c ,

$$\text{adj}(cZ) = c \text{adj}(Z). \quad (3.20)$$

Therefore

$$\det(cZ) = c^2 \det(Z). \quad (3.21)$$

Whenever $\det(Z) \neq 0$, define

$$Z^{-1} := \frac{\text{adj}(Z)}{\det(Z)} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2} \quad (3.22)$$

$$Z^{-1}Z = ZZ^{-1} = 1. \quad (3.23)$$

For another paravector $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, division by Z is generally order-sensitive and must be written using an explicitly ordered inverse. For invertible Z , the unambiguous quotient is defined when Z' and Z commute:

$$\frac{Z'}{Z} := Z'Z^{-1} = Z^{-1}Z' \quad \text{if} \quad \mathbf{V} \times \mathbf{V}' = 0. \quad (3.24)$$

Differentiating $Z^{-1}Z = 1$ gives the general inverse rule

$$d(Z^{-1}) = -Z^{-1} dZ Z^{-1}. \quad (3.25)$$

Both trace and determinant are cyclic:

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (3.26)$$

$$\det(ZZ') = \det(Z'Z). \quad (3.27)$$

For three factors,

$$\text{tr}(ZZ'Z'') = \text{tr}(Z'Z''Z) = \text{tr}(Z''Z'Z), \quad (3.28)$$

$$\det(ZZ'Z'') = \det(Z'Z''Z) = \det(Z''Z'Z). \quad (3.29)$$

For any complex paravector Z and invertible complex paravector T , let

$$Z' := T^{-1}ZT. \quad (3.30)$$

Cyclicity then gives

$$\text{tr}(Z') = \text{tr}(T^{-1}ZT) = \text{tr}(ZTT^{-1}) = \text{tr}(Z), \quad (3.31)$$

$$\det(Z') = \det(T^{-1}ZT) = \det(ZTT^{-1}) = \det(Z). \quad (3.32)$$

Replacing T by T^{-1} gives the same two invariances for $Z' = TZT^{-1}$. Finally, define real and imaginary paravector extraction by

$$\text{re}(Z) := \frac{1}{2}(Z + Z^{\text{H}}) = X, \quad (3.33)$$

$$\text{im}(Z) := \frac{Z - Z^{\text{H}}}{2i} = Y. \quad (3.34)$$

The functions re and im also act componentwise on complex scalars and vectors. The scalar and vector parts are recovered by

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{\text{H}}), \quad (3.35)$$

$$\mathbf{V} = \text{vec}(Z), \quad (3.36)$$

$$\mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{\text{H}}). \quad (3.37)$$

The four real coefficients then follow by composition:

$$a = \text{re}(S) = \text{sc}(X) = \text{sc}(\text{re}(Z)), \quad (3.38)$$

$$b = \text{im}(S) = \text{sc}(Y) = \text{sc}(\text{im}(Z)), \quad (3.39)$$

$$\mathbf{A} = \text{re}(\mathbf{V}) = \text{vec}(X) = \text{vec}(\text{re}(Z)), \quad (3.40)$$

$$\mathbf{B} = \text{im}(\mathbf{V}) = \text{vec}(Y) = \text{vec}(\text{im}(Z)). \quad (3.41)$$

These definitions are intrinsic to the algebra. Appendix A later reproduces them through the faithful Pauli-matrix representation.

3.2 Products and quadratic forms

From the ordered sigma multiplication rule,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \quad (3.42)$$

$$= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 A_n B_m \epsilon_{nmk} i \sigma_k \quad (3.43)$$

$$= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.44)$$

Adding the product with $\mathbf{A}$ and $\mathbf{B}$ interchanged gives

$$2(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}). \quad (3.45)$$

Consequently, for $z^2 \neq 1$ where the inverse occurs,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^2 = \mathbf{A}^2, \quad (3.46)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{A}^2)^n, \quad (3.47)$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} = (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (3.48)$$

Also,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}. \quad (3.49)$$

The corresponding commutator is

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.50)$$

Hence

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] = -2i\mathbf{A} \times \boldsymbol{\sigma}, \quad (3.51)$$

$$[\mathbf{A} \cdot i\boldsymbol{\sigma}, \boldsymbol{\sigma}] = 2\mathbf{A} \times \boldsymbol{\sigma}, \quad (3.52)$$

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.53)$$

3.3 Quadratic identity and eigenvalues

Since $Z - S = \mathbf{V} \cdot \boldsymbol{\sigma}$ and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$,

$$(Z - S)^2 = \mathbf{V}^2, \quad (3.54)$$

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0. \quad (3.55)$$

Using $\text{tr}(Z) = 2S$ and $\det(Z) = S^2 - \mathbf{V}^2$ gives the equivalent trace–determinant identity

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (3.56)$$

Replacing Z by a scalar eigenvalue λ and solving this quadratic gives

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = S \pm \sqrt{\mathbf{V}^2}. \quad (3.57)$$

Because $\mathbf{V}$ is generally complex, $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0. \quad (3.58)$$

3.4 Real products

Let

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \quad (3.59)$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.60)$$

Here $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$. Then

$$XY = (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.61)$$

$$\text{sc}(XY) = ab + \mathbf{A} \cdot \mathbf{B}, \quad (3.62)$$

$$\text{vec}(XY) = a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}, \quad (3.63)$$

$$[X, Y] = 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.64)$$

$$X^2 = (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \quad (3.65)$$

$$\text{sc}(X^2) = a^2 + \mathbf{A}^2, \quad (3.66)$$

$$\text{vec}(X^2) = 2a\mathbf{A}, \quad (3.67)$$

$$X^2 - 2aX + (a^2 - \mathbf{A}^2) = 0. \quad (3.68)$$

The condition $\mathbf{A} \times \mathbf{B} = 0$ implies $XY = YX$ and $X^{-1}Y = YX^{-1}$.

3.5 Complex products

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$,

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.69)$$

Thus

$$\text{sc}(ZZ') = SS' + \mathbf{V} \cdot \mathbf{V}', \quad (3.70)$$

$$\text{vec}(ZZ') = S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}'. \quad (3.71)$$

Equivalently, if $Z = X + \mathbf{i}Y$ and $Z' = X' + \mathbf{i}Y'$, then

$$ZZ' = (XX' - YY') + \mathbf{i}(XY' + YX'). \quad (3.72)$$

Let

$$Z = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (3.73)$$

where $a, b \in \mathbb{R}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, and use primes for a second paravector with real primed components. The four real components of ZZ' are

$$\text{real scalar: } aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \quad (3.74)$$

$$\text{imaginary scalar: } ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \quad (3.75)$$

$$\text{real vector: } a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \quad (3.76)$$

$$\text{imaginary vector: } a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \quad (3.77)$$

For the important quadratic products:

$$Z^2 = (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2i(ab + \mathbf{A} \cdot \mathbf{B}) + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.78)$$

$$ZZ^* = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) - 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.79)$$

$$Z^*Z = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) + 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2i(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.80)$$

$$ZZ^H = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.81)$$

$$Z^H Z = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.82)$$

$$ZZ^{*H} = \det(Z) = a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (3.83)$$

The scalar part is cyclic, $\text{sc}(ZZ') = \text{sc}(Z'Z)$. The operator laws already established also give $(ZZ')^{*H} = Z'^{*H}Z^{*H}$. The opposite cross-product signs follow the displayed product order. Thus $ZZ^* = Z^*Z$ and $ZZ^H = Z^H Z$ precisely when $\mathbf{A} \times \mathbf{B} = 0$.

3.5.1 Norms

For $S = a + bi$ and $\mathbf{V} = \mathbf{A} + \mathbf{B}i$, with real components $a, b, \mathbf{A}, \mathbf{B}$, define

$$\|S\|^2 := SS^* = S^*S = a^2 + b^2, \quad (3.84)$$

$$\|\mathbf{V}\|^2 := \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} = \mathbf{A}^2 + \mathbf{B}^2, \quad (3.85)$$

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2. \quad (3.86)$$

Equivalently,

$$\|Z\|^2 = \text{sc}(Z^H Z) = \frac{1}{2} \text{tr}(Z^H Z) = \text{sc}(ZZ^H) = \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*). \quad (3.87)$$

These norms are not algebraic squares; in general,

$$S^2 \neq \|S\|^2 \quad (3.88)$$

$$\mathbf{V}^2 \neq \|\mathbf{V}\|^2. \quad (3.89)$$

For any complex scalar c ,

$$\|cZ\|^2 = \|c\|^2 \|Z\|^2. \quad (3.90)$$

3.5.2 Comparison with the determinant

The norm is real and nonnegative, whereas $\det(Z)$ may be complex. For $c = re^{i\theta}$ with real r and θ , the determinant retains the phase while the norm does not:

$$\det(re^{i\theta} Z) = r^2 \det(Z) e^{2i\theta}, \quad (3.91)$$

$$\|re^{i\theta} Z\|^2 = r^2 \|Z\|^2. \quad (3.92)$$

3.6 Projection operators

For a real unit vector $\mathbf{u}$, define

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.93)$$

Since $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$,

$$\Pi_{\pm}^2 = \Pi_{\pm}, \quad (3.94)$$

$$\Pi_+ \Pi_- = 0, \quad (3.95)$$

$$\Pi_+ + \Pi_- = 1, \quad (3.96)$$

$$\Pi_{\pm}^H = \Pi_{\pm}, \quad (3.97)$$

$$\Pi_{\pm}^* = \Pi_{\mp}, \quad (3.98)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} = \pm \Pi_{\pm}. \quad (3.99)$$

Thus $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. For two unit directions,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.100)$$

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.101)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (3.102)$$

Define $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$ and $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$. Then

$$\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \quad (3.103)$$

$$\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} = \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \Pi_{\mp}. \quad (3.104)$$

For comparison, define $\Pi'_{\pm} := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})$. Their products are

$$\Pi_+ \Pi'_+ = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} + \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.105)$$

$$\Pi_+ \Pi'_- = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\mathbf{u} - \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.106)$$

$$\Pi_- \Pi'_+ = \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} + \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \quad (3.107)$$

$$\Pi_- \Pi'_- = \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(-\mathbf{u} - \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \quad (3.108)$$

3.6.1 Ladder operators

For projections along the third axis, define

$$Z_{\pm} = \frac{1}{2}(\sigma_1 \pm i\sigma_2) \quad (3.109)$$

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \sigma_3). \quad (3.110)$$

Then

$$[\sigma_3, Z_{\pm}] = \pm 2Z_{\pm}, \quad (3.111)$$

$$Z_{\pm}^2 = 0, \quad (3.112)$$

$$Z_{\pm} Z_{\mp} = \Pi_{\pm}, \quad (3.113)$$

$$\sigma_1 = Z_+ + Z_-, \quad (3.114)$$

$$\sigma_2 = (Z_+ - Z_-)/i, \quad (3.115)$$

$$\sigma_3 = \Pi_+ - \Pi_-. \quad (3.116)$$

The ladder operators map between the two projected sectors according to

$$Z_+ \Pi_+ = 0, \quad (3.117)$$

$$Z_+ \Pi_- = Z_+, \quad (3.118)$$

$$\Pi_+ Z_+ = Z_+, \quad (3.119)$$

$$\Pi_- Z_+ = 0, \quad (3.120)$$

$$Z_- \Pi_- = 0, \quad (3.121)$$

$$Z_- \Pi_+ = Z_-, \quad (3.122)$$

$$\Pi_- Z_- = Z_-, \quad (3.123)$$

$$\Pi_+ Z_- = 0. \quad (3.124)$$

3.7 Functions of paravectors

3.7.1 General projector rule

Let $\lambda_+, \lambda_- \in \mathbb{C}$. Suppose $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+ + \Pi_- = 1$, $\Pi_+ \Pi_- = \Pi_- \Pi_+ = 0$, and

$$Z = \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.125)$$

Then orthogonality gives

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_-. \quad (3.126)$$

Thus, for a convergent power series $f(w) = \sum_{n=0}^{\infty} c_n w^n$,

$$f(Z) = \sum_{n=0}^{\infty} c_n Z^n \quad (3.127)$$

$$= f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.128)$$

3.7.2 Functions of a real unit vector

For a real unit vector $\mathbf{u}$, let $X = \mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$. Its eigenvalues are ± 1 . The general rule and $\Pi_\pm = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$ give

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(1) \Pi_+ + f(-1) \Pi_- = \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1)) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.129)$$

For the square root, the independent scalar choices $\sqrt{1} = \pm 1$ and $\sqrt{-1} = \pm i$ give

$$\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} = \pm \Pi_+ \pm i \Pi_-. \quad (3.130)$$

3.7.3 Real paravectors

Let $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$, where $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$. For $\mathbf{A} \neq 0$, put

$$\mathbf{u} := \frac{\mathbf{A}}{\|\mathbf{A}\|} \quad (3.131)$$

$$w := \|\mathbf{A}\|. \quad (3.132)$$

Then define

$$\lambda_{\pm} := a \pm w. \quad (3.133)$$

Using $1 = \Pi_+ + \Pi_-$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$ gives

$$X = a(\Pi_+ + \Pi_-) + w(\Pi_+ - \Pi_-) \quad (3.134)$$

$$= \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.135)$$

The general rule therefore gives

$$f(X) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.136)$$

With nonzero eigenvalues where required and chosen function branches, this includes

$$X^n = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.137)$$

$$X^{-1} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-, \quad (3.138)$$

$$\sqrt{X} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-. \quad (3.139)$$

The exponential and logarithm are developed below. The spectral inverse agrees with the adjugate–determinant formula developed earlier. Indeed,

$$\text{adj}(X) = \lambda_- \Pi_+ + \lambda_+ \Pi_- \quad (3.140)$$

$$\det(X) = \lambda_+ \lambda_-, \quad (3.141)$$

and therefore, when $\lambda_+ \lambda_- \neq 0$,

$$\frac{\text{adj}(X)}{\det(X)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_- = X^{-1}. \quad (3.142)$$

The formulas extend continuously to $\mathbf{A} = 0$.

3.7.4 Direct square-root components

Let $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, where $b \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$. If $X^2 = Y$, then

$$a^2 = \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right) \quad (3.143)$$

$$\mathbf{A}^2 = b - a^2, \quad (3.144)$$

with signs and vector direction chosen consistently.

3.7.5 Complex spectral form

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with $S \in \mathbb{C}$, $\mathbf{V} \in \mathbb{C}^3$, and $\mathbf{V}^2 \neq 0$, define

$$w := +\sqrt{\mathbf{V}^2}. \quad (3.145)$$

Here the square root is the principal scalar square root, so $w^2 = \mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$. Define

$$\mathbf{u} := \frac{\mathbf{V}}{w}. \quad (3.146)$$

This normalization uses the complex bilinear dot product:

$$\mathbf{u} \cdot \mathbf{u} = \frac{\mathbf{V} \cdot \mathbf{V}}{w^2} = 1. \quad (3.147)$$

It does not generally normalize the Hermitian norm, since

$$\|\mathbf{u}\|^2 = \mathbf{u}^* \cdot \mathbf{u} = \frac{\|\mathbf{V}\|^2}{\|w\|^2}. \quad (3.148)$$

This quantity need not equal 1. Thus $\mathbf{u}$ is a complex bilinear unit direction, not generally a unit vector in the norm defined above. Define

$$\lambda_{\pm} := S \pm w. \quad (3.149)$$

Define

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.150)$$

Because $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = \mathbf{u} \cdot \mathbf{u} = 1$,

$$\Pi_{\pm}^2 = \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = \Pi_{\pm}, \quad (3.151)$$

$$\Pi_+ \Pi_- = \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0. \quad (3.152)$$

Also, $1 = \Pi_+ + \Pi_-$ and $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$, so

$$Z = S(\Pi_+ + \Pi_-) + w(\Pi_+ - \Pi_-) \quad (3.153)$$

$$= \lambda_+ \Pi_+ + \lambda_- \Pi_-. \quad (3.154)$$

The general rule gives

$$f(Z) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.155)$$

The adjugate and determinant have the corresponding spectral forms

$$\text{adj}(Z) = \lambda_- \Pi_+ + \lambda_+ \Pi_- \quad (3.156)$$

$$\det(Z) = \lambda_+ \lambda_-. \quad (3.157)$$

Consequently, when $\lambda_+ \lambda_- \neq 0$,

$$\frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_- = Z^{-1}. \quad (3.158)$$

Let $\sqrt{\lambda_+}$ and $\sqrt{\lambda_-}$ denote the principal scalar roots. The two signs below are chosen independently, giving the four square roots

$$\sqrt{Z} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-. \quad (3.159)$$

3.7.6 Exponential of a general complex paravector

Let $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, with w and $\mathbf{u} = \mathbf{V}/w$ as defined above.

3.7.7 Projection derivation

The projected function rule (3.155), with $\lambda_{\pm} = S \pm w$ and $\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$, gives

$$\exp(Z) = \frac{\exp(S + w)}{2}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \frac{\exp(S - w)}{2}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.160)$$

Factoring out $\exp(S)$ and collecting the scalar and vector parts gives

$$\exp(Z) = \exp(S) \left(\frac{\exp(w) + \exp(-w)}{2} + \frac{\exp(w) - \exp(-w)}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.161)$$

3.7.8 Direct series derivation

Because S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$,

$$\exp(Z) = \exp(S) \sum_{n=0}^{\infty} \frac{(w \mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!}. \quad (3.162)$$

The identity $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ gives, for every nonnegative integer n ,

$$(w \mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = w^{2n} \quad (3.163)$$

$$(w \mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = w^{2n+1} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.164)$$

Separating the series into even and odd powers therefore gives

$$\exp(Z) = \exp(S) \left(\sum_{n=0}^{\infty} \frac{w^{2n}}{(2n)!} + \left(\sum_{n=0}^{\infty} \frac{w^{2n+1}}{(2n+1)!} \right) \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.165)$$

The scalar definitions of $\cosh w$ and $\sinh w$ reduce both derivations to the same general result:

$$\exp(Z) = \exp(S)(\cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.166)$$

This result does not depend on which root is chosen. Explicitly,

$$\cosh(-w) = \cosh w \quad (3.167)$$

$$\sinh(-w)(-\mathbf{u}) = \sinh w \mathbf{u}. \quad (3.168)$$

Replacing the principal root w by the other algebraic root $-w$ also changes $\mathbf{u} = \mathbf{V}/w$ to $-\mathbf{u}$. Thus both terms in the general result are unchanged.

3.7.9 Component form and real/imaginary limits

For

$$Z = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (3.169)$$

the principal root is

$$w = +\sqrt{\mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}}. \quad (3.170)$$

The corresponding complex bilinear unit direction is

$$\mathbf{u} = \frac{\mathbf{A} + \mathbf{B}i}{w}. \quad (3.171)$$

The fully expanded formula is

$$\exp(Z) = \exp(a)(\cos b + i \sin b) \left(\cosh w + \frac{\sinh w}{w} (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma} \right). \quad (3.172)$$

For a real vector, choose $w = \|\mathbf{A}\|$. Then

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \cosh \|\mathbf{A}\| + \sinh \|\mathbf{A}\| \frac{\mathbf{A}}{\|\mathbf{A}\|} \cdot \boldsymbol{\sigma}. \quad (3.173)$$

Because the real scalar a commutes with the vector part, the strictly real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ satisfies

$$\exp(X) = \exp(a) \left(\cosh \|\mathbf{A}\| + \sinh \|\mathbf{A}\| \frac{\mathbf{A}}{\|\mathbf{A}\|} \cdot \boldsymbol{\sigma} \right). \quad (3.174)$$

For an imaginary vector, choose $w = i\|\mathbf{B}\|$. Then

$$\exp(i\mathbf{B} \cdot \boldsymbol{\sigma}) = \cos \|\mathbf{B}\| + i \sin \|\mathbf{B}\| \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.175)$$

Because the imaginary scalar ib also commutes with the vector part, the strictly imaginary paravector $Z = i(b + \mathbf{B} \cdot \boldsymbol{\sigma})$ satisfies

$$\exp(Z) = (\cos b + i \sin b) \left(\cos \|\mathbf{B}\| + i \sin \|\mathbf{B}\| \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma} \right). \quad (3.176)$$

Each quotient in these formulas is understood by its analytic limit at a zero vector.

3.7.10 Kinematic form

The real-vector exponential has a direct kinematic interpretation. Define

$$\gamma := \cosh \|\mathbf{A}\| \quad (3.177)$$

$$\mathbf{v} := \tanh \|\mathbf{A}\| \frac{\mathbf{A}}{\|\mathbf{A}\|}. \quad (3.178)$$

Then $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ and

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (3.179)$$

Thus $\|\mathbf{A}\|$ is the rapidity and $\mathbf{A}/\|\mathbf{A}\|$ is the direction of motion. This is the normalized future-directed paravector used below for four-velocity.

3.7.11 Cayley transform

For the real paravector $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, assume $0 < \|\mathbf{B}\| < b$ and put

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|}. \quad (3.180)$$

Because Y is real,

$$Y^* = \text{adj}(Y) = b - \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.181)$$

Its determinant is

$$\det(Y) = Y \operatorname{adj}(Y) = b^2 - \|\mathbf{B}\|^2. \quad (3.182)$$

The assumed range makes $\operatorname{adj}(Y)$ invertible. Its vector part is collinear with that of Y , so the unambiguous quotient notation introduced above applies. Multiplication by Y/Y gives

$$\frac{Y}{\operatorname{adj}(Y)} = \frac{Y}{\operatorname{adj}(Y)} \frac{Y}{Y} = \frac{Y^2}{\det(Y)}. \quad (3.183)$$

The numerator on the right expands as

$$Y^2 = (b + \mathbf{B} \cdot \boldsymbol{\sigma})^2 = b^2 + \|\mathbf{B}\|^2 + 2b\mathbf{B} \cdot \boldsymbol{\sigma}. \quad (3.184)$$

Dividing this expression by $\det(Y)$ therefore gives

$$\frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} = \frac{b^2 + \|\mathbf{B}\|^2}{b^2 - \|\mathbf{B}\|^2} + \frac{2b\|\mathbf{B}\|}{b^2 - \|\mathbf{B}\|^2} \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.185)$$

Define

$$\theta := 2 \tanh^{-1} \left(\frac{\|\mathbf{B}\|}{b} \right). \quad (3.186)$$

The double-angle identities give

$$\cosh \theta = \frac{b^2 + \|\mathbf{B}\|^2}{b^2 - \|\mathbf{B}\|^2} \quad (3.187)$$

$$\sinh \theta = \frac{2b\|\mathbf{B}\|}{b^2 - \|\mathbf{B}\|^2}. \quad (3.188)$$

Therefore

$$\begin{aligned} \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\theta) \Pi_+ + \exp(-\theta) \Pi_-. \end{aligned} \quad (3.189)$$

Comparing this result with the kinematic form above identifies its boost-generator vector:

$$\mathbf{A} = \theta \mathbf{u} \quad (3.190)$$

$$\theta = \|\mathbf{A}\|. \quad (3.191)$$

The definition of θ then gives the map from the boost-generator vector $\mathbf{A}$ to the Cayley vector $\mathbf{B}$:

$$\mathbf{B} = b \tanh \left(\frac{\|\mathbf{A}\|}{2} \right) \frac{\mathbf{A}}{\|\mathbf{A}\|}. \quad (3.192)$$

Thus $\mathbf{A}$ is the rapidity vector, whereas $\mathbf{B}/b$ is the Cayley parameter; its magnitude is the hyperbolic tangent of half the rapidity. The choice $b = 1$ recovers the normalized Cayley form, and the case $\mathbf{B} = 0$ follows by continuity.

Finally, the double-angle identities express the kinematic quantities directly in terms of the Cayley variables:

$$\gamma = \frac{b^2 + \|\mathbf{B}\|^2}{b^2 - \|\mathbf{B}\|^2} \quad (3.193)$$

$$\mathbf{v} = \frac{2b\mathbf{B}}{b^2 - \|\mathbf{B}\|^2}. \quad (3.194)$$

Substitution makes the three equivalent forms explicit:

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}}. \quad (3.195)$$

3.7.12 Aligned boost and rotation

When $\mathbf{u}$ is real, write w in terms of real ϕ and θ as

$$w := \phi + i\theta. \quad (3.196)$$

The parallel real and imaginary generators commute, so

$$\exp(w\mathbf{u} \cdot \boldsymbol{\sigma}) = \exp(\phi\mathbf{u} \cdot \boldsymbol{\sigma}) \exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.197)$$

The factors are

$$\exp(\phi\mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh \phi + \sinh \phi \mathbf{u} \cdot \boldsymbol{\sigma} \quad (3.198)$$

$$\exp(i\theta\mathbf{u} \cdot \boldsymbol{\sigma}) = \cos \theta + i \sin \theta \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.199)$$

Thus ϕ is boost-like and θ is rotation-like. Their product is

$$\exp(w\mathbf{u} \cdot \boldsymbol{\sigma}) = (\cosh \phi \cos \theta + i \sinh \phi \sin \theta) + (\sinh \phi \cos \theta + i \cosh \phi \sin \theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.200)$$

With the conventional factor $1/2$ in the transformation generator, 2ϕ is the rapidity and 2θ is the rotation angle. For complex $\mathbf{u}$, the real and imaginary generator vectors need not be parallel, so this interpretation does not generally apply.

If $\mathbf{V}^2 = 0$, then $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = 0$, even when $\mathbf{V}$ is a nonzero complex-null vector, and the exponential terminates:

$$\exp(Z) = \exp(S)(1 + \mathbf{V} \cdot \boldsymbol{\sigma}). \quad (3.201)$$

Finally, the defining power series respects both conjugation operations:

$$\exp(Z^*) = \exp(Z)^* \quad (3.202)$$

$$\exp(Z^H) = \exp(Z)^H. \quad (3.203)$$

3.7.13 Logarithm of a general complex paravector

For a nonzero complex scalar λ , each scalar logarithm has the form

$$\log \lambda = \log \|\lambda\| + i(\arg \lambda + 2n\pi), \quad n \in \mathbb{Z}. \quad (3.204)$$

For the complex paravector and spectral quantities defined above, assume both eigenvalues are nonzero and choose one scalar logarithm of each. The projected function rule gives

$$\log Z = \log(\lambda_+) \Pi_+ + \log(\lambda_-) \Pi_-. \quad (3.205)$$

Substitution of the projectors and collection of the scalar and vector parts gives

$$\log Z = \frac{1}{2} (\log(\lambda_+) + \log(\lambda_-)) + \frac{1}{2} (\log(\lambda_+) - \log(\lambda_-)) \mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.206)$$

Changing w to $-w$ changes $\mathbf{u}$ to $-\mathbf{u}$ and interchanges λ_+ with λ_- and Π_+ with Π_- , leaving this result unchanged. Independent branch choices for the two scalar logarithms give the different logarithms of Z .

3.7.14 Special cases

For a real unit vector $\mathbf{u}$, the paravector $\mathbf{u} \cdot \boldsymbol{\sigma}$ has eigenvalues 1 and -1 . Hence all projected branches are

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i \Pi_+ + (2m+1)\pi i \Pi_-, \quad n, m \in \mathbb{Z}. \quad (3.207)$$

For the positive real paravector $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, with $b > \|\mathbf{B}\|$, the real logarithm is

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \tanh^{-1} \left(\frac{\|\mathbf{B}\|}{b} \right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}. \quad (3.208)$$

The vector quotient is understood by its limit when $\mathbf{B} = 0$.

The square-root and logarithm formulas above are likewise understood by analytic continuation wherever their chosen branches are regular.

3.8 Differentiated fields

For products of fields,

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z'. \quad (3.209)$$

Let $F = (a + bi) + (\mathbf{A} + \mathbf{B}i) \cdot \boldsymbol{\sigma}$ depend on $(t, \mathbf{r})$. Then

$$\partial F = (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + i(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \quad (3.210)$$

$$+ (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \quad (3.211)$$

$$+ i(\partial_{\mathbf{r}} b + \partial_{\mathbf{r}} \times \mathbf{A} + \partial_t \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.212)$$

Multiplication by i rotates these four components: the real scalar becomes $-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$, the imaginary scalar becomes $\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$, the real vector becomes the negative of the imaginary vector above, and the imaginary vector becomes the real vector.

4 Transformations and invariants

The transformations used below are organized by the quantities they leave unchanged:

$$\text{tr}(Z) = 2 \text{sc}(Z) \quad (4.1)$$

$$\det(Z) \quad (4.2)$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(Z^{\mathbf{H}} Z). \quad (4.3)$$

The trace is linear; the determinant and norm are quadratic. Here T and T' denote general complex paravectors, with T' a second transformation factor rather than a derivative. The symbol U denotes a complex unitary vector:

$$U^{\mathbf{H}} U = U U^{\mathbf{H}} = 1 \quad (4.4)$$

$$U^{\mathbf{H}} = U^{-1}. \quad (4.5)$$

Here U is a transformation; Section 5 later reuses the symbol for four-velocity.

4.1 Similarity invariants

4.1.1 Trace invariance

For invertible T , consider either similarity:

$$Z' = T^{-1} Z T, \quad (4.6)$$

$$Z' = T Z T^{-1}. \quad (4.7)$$

For example, cyclicity gives

$$\text{tr}(T^{-1} Z T) = \text{tr}(Z T T^{-1}) = \text{tr}(Z). \quad (4.8)$$

Thus $\text{tr}(Z') = \text{tr}(Z)$. For unitary U , the same similarities are

$$Z' = U^H Z U, \quad (4.9)$$

$$Z' = U Z U^H. \quad (4.10)$$

4.1.2 Determinant invariance

For the general two-sided transformation $Z' = T' Z T$,

$$\det(Z') = \det(T') \det(T) \det(Z). \quad (4.11)$$

Hence $\det(T') \det(T) = 1$ guarantees $\det(Z') = \det(Z)$, and is required if this equality is to hold for arbitrary Z . Equivalently, when $\det(T) \neq 0$, any T' satisfying

$$\det(T') = \frac{1}{\det(T)} \quad (4.12)$$

preserves the determinant of every Z . The automatic choice $T' = T^{-1}$ gives the similarity

$$Z' = T^{-1} Z T. \quad (4.13)$$

The opposite ordering $Z' = T Z T^{-1}$ preserves the determinant as well. For unitary U ,

$$\det(U^H) \det(U) = \det(U^{-1}) \det(U) = 1, \quad (4.14)$$

so both unitary sandwiches preserve the determinant.

4.1.3 Norm invariance

Norm invariance requires the unitary condition (4.4). For example, if $Z' = U^H Z U$, then

$$\|Z'\|^2 = \frac{1}{2} \text{tr}(U^H Z^H U U^H Z U) \quad (4.15)$$

$$= \frac{1}{2} \text{tr}(Z^H Z) = \|Z\|^2. \quad (4.16)$$

The same equality $\|Z'\|^2 = \|Z\|^2$ holds for

$$Z' = U Z \quad (4.17)$$

$$Z' = Z U \quad (4.18)$$

$$Z' = U^H Z U \quad (4.19)$$

$$Z' = U Z U^H. \quad (4.20)$$

It does not hold in general when T is nonunitary.

Rotations and boosts preserve different combinations of these invariants. A rotation satisfies

$$R^H = R^{-1} \quad (4.21)$$

$$R^H R = R R^H = 1, \quad (4.22)$$

so R is unitary; $Z' = R^H Z R$ preserves trace, determinant, and norm. A nonzero boost satisfies

$$L^H = L \quad (4.23)$$

$$\det(L) = 1 \quad (4.24)$$

$$L^H L = L^2 \neq 1, \quad (4.25)$$

so L is not unitary. The transformation $Z' = L^{-1} Z L^{-1}$ preserves the determinant but does not generally preserve the trace or norm.

5 Space–time kinematics

5.1 Intervals, velocity, and momentum

Represent a differential event by the real paravector

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (5.1)$$

$$dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}. \quad (5.2)$$

The Minkowski interval is

$$ds^2 = \det(dX) = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (5.3)$$

For any invertible complex paravector T , define

$$dX' := T^{-1} dX T. \quad (5.4)$$

The cyclic determinant property gives

$$\det(dX') = \det(T^{-1} dX T) = \det(dX T T^{-1}) = \det(dX) = ds^2. \quad (5.5)$$

Thus similarity by any such T leaves the determinant unchanged. This is an algebraic change of basis; even when $T = L$, it is not the physical boost of a space–time paravector. The latter is the congruence $dX' = L^{-1} dX L^{-1}$ developed in Section 8.2. The alternative similarity ordering $T dX T^{-1}$ has the same determinant invariance. For a future-directed time-like interval, choose the positive scalar root

$$ds := \sqrt{\det(dX)}. \quad (5.6)$$

The sign of the invariant squared interval classifies the causal relation between the two neighboring events:

$$ds^2 > 0 : \quad \text{time-like; a massive particle can follow the interval,} \quad (5.7)$$

$$ds^2 = 0 : \quad \text{null; light can follow the interval,} \quad (5.8)$$

$$ds^2 < 0 : \quad \text{space-like; no causal signal can follow the interval.} \quad (5.9)$$

Only in the time-like case is ds real and nonzero under the chosen positive-root convention; along a time-like worldline it is the proper-time differential. Thus the following kinematics explores the determinant of dX and its normalization by this scalar. For two intervals,

$$[dX, dX'] = 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \quad (5.10)$$

When $ds^2 \neq 0$,

$$dX^{-1} = \frac{\text{adj}(dX)}{\det(dX)} = \frac{dX^*}{ds^2}. \quad (5.11)$$

A null interval has $dt^2 = d\mathbf{r}^2$ and no inverse. Along a future-directed time-like worldline, s is proper time. Define the coordinate velocity by

$$\mathbf{v} := \frac{d\mathbf{r}}{dt}. \quad (5.12)$$

Then $d\mathbf{r} = \mathbf{v} dt$, and the positive proper-time increment satisfies

$$ds^2 = (1 - \mathbf{v}^2) dt^2 \quad (5.13)$$

$$ds = \sqrt{1 - \mathbf{v}^2} dt \quad (5.14)$$

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} \quad (5.15)$$

$$ds = \frac{dt}{\gamma} \quad (5.16)$$

$$\gamma = \frac{dt}{ds}. \quad (5.17)$$

For any quantity evaluated along the worldline, the one-variable chain rule therefore gives

$$\frac{d}{ds} = \frac{dt}{ds} \frac{d}{dt} = \gamma \frac{d}{dt}. \quad (5.18)$$

Applying this operator a second time gives

$$\frac{d^2}{ds^2} = \gamma \frac{d}{dt} \left(\gamma \frac{d}{dt} \right) = \gamma^2 \frac{d^2}{dt^2} + \gamma \frac{d\gamma}{dt} \frac{d}{dt}. \quad (5.19)$$

In particular, the proper velocity is

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (5.20)$$

Using the scalar-scaling rule (3.21), its determinant is

$$\det(U) = \det \left(\frac{dX}{ds} \right) = \frac{\det(dX)}{ds^2} = 1. \quad (5.21)$$

Equivalently, $\det(U) = \gamma^2(1 - \mathbf{v}^2) = 1$, so

$$U^{-1} = \frac{\text{adj}(U)}{\det(U)} = U^*. \quad (5.22)$$

Proper time therefore obeys

$$\tau = \int ds = \int \sqrt{1 - \mathbf{v}^2} dt = \int \frac{dt}{\gamma}. \quad (5.23)$$

The four-momentum is the paravector

$$mU = m\gamma + m\gamma\mathbf{v} \cdot \boldsymbol{\sigma} = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (5.24)$$

Its scalar and vector components define

$$E := m\gamma \quad (5.25)$$

$$\mathbf{P} := m\gamma\mathbf{v}. \quad (5.26)$$

Taking its determinant gives the mass-shell relation

$$\det(mU) = m^2 \det(U) = m^2 = \det(E + \mathbf{P} \cdot \boldsymbol{\sigma}) = E^2 - \mathbf{P}^2, \quad (5.27)$$

or $E^2 = \mathbf{P}^2 + m^2$; for $m = 0$, $E^2 = \mathbf{P}^2$.

5.1.1 Proper acceleration

The required worldline quantities are

$$\mathbf{v} := \frac{d\mathbf{r}}{dt} \quad (5.28)$$

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} \quad (5.29)$$

$$\gamma := \frac{dt}{ds} \quad (5.30)$$

$$U := \frac{dX}{ds}. \quad (5.31)$$

Differentiating $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ gives

$$\begin{aligned} \frac{d\gamma}{dt} &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2} \frac{d(1 - \mathbf{v}^2)}{dt} \\ &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2} (-2\mathbf{v} \cdot \mathbf{a}) \\ &= \gamma^3 \mathbf{v} \cdot \mathbf{a}. \end{aligned} \quad (5.32)$$

Hence the second proper-time derivative is

$$\frac{d^2}{ds^2} = \gamma^2 \frac{d^2}{dt^2} + \gamma^4 (\mathbf{v} \cdot \mathbf{a}) \frac{d}{dt}. \quad (5.33)$$

Using $U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ gives the proper-acceleration paravector

$$\begin{aligned} \frac{dU}{ds} &= \frac{d^2 X}{ds^2} = \gamma \frac{d}{dt} (\gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \\ &= \gamma \frac{d\gamma}{dt} (1 + \mathbf{v} \cdot \boldsymbol{\sigma}) + \gamma^2 \mathbf{a} \cdot \boldsymbol{\sigma} \\ &= \gamma^4 (\mathbf{v} \cdot \mathbf{a}) + (\gamma^4 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (5.34)$$

Differentiate the normalization $U^*U = 1$:

$$0 = \frac{d(U^*U)}{ds} = \left(\frac{dU}{ds} \right)^* U + U^* \frac{dU}{ds}. \quad (5.35)$$

Sigma conjugation leaves the scalar part unchanged, and the scalar part is cyclic, so the two scalar terms are equal. Therefore

$$\text{sc}(U^* dU/ds) = 0. \quad (5.36)$$

This is the paravector statement that four-velocity and four-acceleration are Minkowski-orthogonal. Since $U^{-1} = U^*$, right multiplication of the differentiated normalization by U^* gives

$$\text{adj}\left(\frac{dU}{ds}\right) = \left(\frac{dU}{ds}\right)^* = -U^* \frac{dU}{ds} U^*. \quad (5.37)$$

Here adjugation equals sigma conjugation because dU/ds is a real paravector. Direct multiplication gives the scalar-free form

$$U^* \frac{dU}{ds} = \gamma^3 (\mathbf{a} - i\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}. \quad (5.38)$$

Its real vector contains the coordinate acceleration, while its imaginary vector records the turning of the velocity direction. Since $\mathbf{a} \cdot (\mathbf{v} \times \mathbf{a}) = 0$, its square is scalar. The adjugate identity therefore yields the determinant without a separate component expansion:

$$\begin{aligned} \det\left(\frac{dU}{ds}\right) &= \text{adj}\left(\frac{dU}{ds}\right) \frac{dU}{ds} \\ &= -\left(U^* \frac{dU}{ds}\right)^2 \\ &= -\gamma^6 (\mathbf{a}^2 - (\mathbf{v} \times \mathbf{a})^2) \\ &= -\gamma^4 (\mathbf{a}^2 + \gamma^2 (\mathbf{v} \cdot \mathbf{a})^2). \end{aligned} \quad (5.39)$$

The last step uses $(\mathbf{v} \times \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}^2 - (\mathbf{v} \cdot \mathbf{a})^2$ and $1 - \mathbf{v}^2 = \gamma^{-2}$. Thus the proper acceleration is space-like unless it vanishes, and $-\det(dU/ds)$ is its invariant squared magnitude.

For $\mathbf{v} \neq 0$, resolve the coordinate acceleration parallel and perpendicular to the velocity by

$$\mathbf{u} := \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad (5.40)$$

$$\mathbf{a}_{\parallel} := (\mathbf{a} \cdot \mathbf{u}) \mathbf{u} \quad (5.41)$$

$$\mathbf{a}_{\perp} := \mathbf{a} - \mathbf{a}_{\parallel}. \quad (5.42)$$

Since $(\mathbf{v} \cdot \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}_{\parallel}^2$ and $1 + \gamma^2 \mathbf{v}^2 = \gamma^2$,

$$\det\left(\frac{dU}{ds}\right) = -\gamma^4 (\mathbf{a}_{\perp}^2 + \gamma^2 \mathbf{a}_{\parallel}^2). \quad (5.43)$$

At $\mathbf{v} = 0$, this reduces continuously to $\det(dU/ds) = -\mathbf{a}^2$.

5.2 Varying transformations and second differentials

Along a path, let d denote successive differentiation with respect to its parameter and define

$$dX' = T^{-1} dX T. \quad (5.44)$$

Although $\det(dX') = \det(dX)$ at each point, a varying T adds a term to the second differential. Define

$$d\Theta := T^{-1} dT \quad (5.45)$$

$$dT^{-1} = -d\Theta T^{-1}. \quad (5.46)$$

Direct differentiation gives

$$d^2 X' = d(T^{-1} dX T) \quad (5.47)$$

$$= T^{-1} d^2 X T - d\Theta dX' + dX' d\Theta \quad (5.48)$$

$$= T^{-1} d^2 X T + [dX', d\Theta]. \quad (5.49)$$

Dividing by the corresponding powers of ds , define

$$\Omega := \frac{d\Theta}{ds} = T^{-1} \frac{dT}{ds}, \quad (5.50)$$

so, with $U = dX/ds$ and $U' = dX'/ds$,

$$\frac{dU'}{ds} = T^{-1} \frac{dU}{ds} T + [U', \Omega]. \quad (5.51)$$

For constant T , the connection term vanishes. For varying T , it is the algebraic source of the additional terms in rotating and accelerating frames and is the pathwise prototype of a covariant derivative.

Here dX' is defined by the pointwise transformation of the tangent. When T varies, it is not generally $d(T^{-1}XT)$, which also contains the transformation of X itself. Moreover, a general complex T can make dX' complex even when dX is real, so a physical space–time transformation requires an additional reality-preserving condition.

The corresponding local algebra basis is

$$\sigma'_n := T^{-1} \sigma_n T. \quad (5.52)$$

Similarity preserves all products, so the transformed generators obey the same sigma algebra. If T varies along the path, direct differentiation gives

$$d\sigma'_n = [\sigma'_n, d\Theta] \quad (5.53)$$

$$d\Theta = T^{-1} dT. \quad (5.54)$$

This equation describes the change of a local algebra basis. It should not be identified with a physical Lorentz transformation until the action of T and the condition preserving real space–time paravectors have both been specified. In particular, the complex boost-like terms obtained from a completely general T are connection coefficients, not automatically components of a real physical velocity.

5.3 Free-particle action

The free-particle action and Lagrangian are

$$\int L dt = -m \int ds \quad (5.55)$$

$$L = -m\sqrt{1 - \mathbf{v}^2} = -\frac{m}{\gamma}. \quad (5.56)$$

For a general Lagrangian $L(t, \mathbf{r}, \mathbf{v})$,

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L \quad (5.57)$$

$$H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \quad (5.58)$$

$$\frac{dH}{dt} = -\partial_t L. \quad (5.59)$$

Thus spatial translation invariance conserves canonical momentum and time translation invariance conserves the Hamiltonian. For the free-particle Lagrangian,

$$\partial_{\mathbf{v}} L = m\gamma \mathbf{v}, \quad (5.60)$$

$$H = m\gamma \mathbf{v}^2 + \frac{m}{\gamma} = m\gamma(\mathbf{v}^2 + \gamma^{-2}) = m\gamma. \quad (5.61)$$

At low velocity, the binomial expansions are

$$\sqrt{1 - \mathbf{v}^2} = 1 - \frac{1}{2}\mathbf{v}^2 + O(v^4) \quad (5.62)$$

$$\gamma = 1 + \frac{1}{2}\mathbf{v}^2 + O(v^4). \quad (5.63)$$

Multiplying the first expansion by $-m$ gives

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m + \frac{1}{2}m\mathbf{v}^2 + O(v^4), \quad (5.64)$$

while the second gives

$$H = m\gamma = m + \frac{1}{2}m\mathbf{v}^2 + O(v^4). \quad (5.65)$$

The exact relativistic kinetic energy is therefore

$$K := H - m = m(\gamma - 1), \quad (5.66)$$

and, apart from the constant rest-energy term, both expansions recover the nonrelativistic kinetic energy $\frac{1}{2}m\mathbf{v}^2$.

6 Electromagnetism

6.1 Field equation and potentials

Introduce the paravector differential operator, electromagnetic field, and source

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \quad (6.1)$$

$$F = (\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma} \quad (6.2)$$

$$D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (6.3)$$

All four Maxwell equations are the single equation

$$\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (6.4)$$

Sigma conjugation gives the equivalent equation

$$\partial^* F^* = D. \quad (6.5)$$

Expanding the left side by component gives

$$\partial F = \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i \partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \quad (6.6)$$

$$+ \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (6.7)$$

$$+ \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (6.8)$$

Equating these four components with those of $\rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives, respectively,

$$\text{real scalar: } \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (6.9)$$

$$\text{imaginary scalar: } \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (6.10)$$

$$\text{real vector: } \partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad (6.11)$$

$$\text{imaginary vector: } \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (6.12)$$

The conjugate operator products give

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2 =: \square, \quad (6.13)$$

whereas

$$\partial^2 = \partial_t^2 + \partial_{\mathbf{r}}^2 + 2\partial_t \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma} \neq \square. \quad (6.14)$$

Applying ∂^* to (6.4) gives

$$\square F = \partial^* D^* \quad (6.15)$$

$$= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i \partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}. \quad (6.16)$$

The real-scalar component is charge conservation, and the real- and imaginary-vector components are the electric and magnetic wave equations:

$$\text{real scalar: } \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (6.17)$$

$$\text{real vector: } \square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (6.18)$$

$$\text{imaginary vector: } \square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (6.19)$$

In vacuum, $\rho = 0$ and $\mathbf{J} = 0$, so

$$\square \mathbf{E} = 0 \quad (6.20)$$

$$\square \mathbf{B} = 0. \quad (6.21)$$

For the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$,

$$\partial \Phi = \underbrace{\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}}_{\text{real scalar}} + \underbrace{(\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \quad (6.22)$$

$$+ \underbrace{i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}. \quad (6.23)$$

Define the scalar component

$$S := \text{sc}(\partial \Phi) = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (6.24)$$

Then $\partial\Phi = S + F^*$: its real-vector and imaginary-vector components give

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_r V \quad (6.25)$$

$$\mathbf{B} = \partial_r \times \mathbf{A}. \quad (6.26)$$

Also, $F = \partial^* \Phi^* - S$, so

$$\partial F = \partial \partial^* \Phi^* - \partial S = \square \Phi^* - \partial S \quad (6.27)$$

$$= (\square V - \partial_t S) - (\square \mathbf{A} + \partial_r S) \cdot \boldsymbol{\sigma}. \quad (6.28)$$

Comparing its real-scalar and real-vector components with $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ gives

$$\rho = \square V - \partial_t S \quad (6.29)$$

$$\mathbf{J} = \square \mathbf{A} + \partial_r S. \quad (6.30)$$

For a scalar field λ , the gauge transformation is

$$\Phi' = \Phi + \partial^* \lambda, \quad (6.31)$$

$$\partial \Phi' = \partial \Phi + \partial \partial^* \lambda = \partial \Phi + \square \lambda. \quad (6.32)$$

Because $\square \lambda$ is scalar, only the scalar component changes:

$$S' = S + \square \lambda \quad (6.33)$$

$$\text{vec}(\partial \Phi') = \text{vec}(\partial \Phi). \quad (6.34)$$

Thus F is unchanged. In the Lorenz gauge $S = 0$, equation (6.29) reduces to

$$\rho = \square V \quad (6.35)$$

$$\mathbf{J} = \square \mathbf{A}. \quad (6.36)$$

6.2 Field invariants

Electric and magnetic fields mix under Lorentz transformations, so their individual magnitudes depend on the observer. Two combinations remain the same for every inertial observer and therefore classify the electromagnetic field itself.

For

$$F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (6.37)$$

the scalar part vanishes, and therefore $\text{adj}(F) = -F$. Hence

$$\det(F) = -F^2 \quad (6.38)$$

$$= -(\mathbf{E} + \mathbf{B}i) \cdot (\mathbf{E} + \mathbf{B}i) \quad (6.39)$$

$$= -\mathbf{E}^2 + \mathbf{B}^2 - 2i\mathbf{E} \cdot \mathbf{B}. \quad (6.40)$$

For any invertible real or complex paravector T , let

$$F' = T^{-1} F T. \quad (6.41)$$

The general similarity identity (3.32) gives

$$\det(F') = \det(T^{-1} F T) = \det(F). \quad (6.42)$$

Rotations and boosts are the physical special cases

$$F' = R^{-1}FR = R^HFR \quad (6.43)$$

$$F' = L^{-1}FL = L^*FL, \quad (6.44)$$

because $R^{-1} = R^H$ and $L^{-1} = L^*$. Hence every inertial observer agrees on

$$a = \mathbf{E}^2 - \mathbf{B}^2 = -\operatorname{re}(\det(F)), \quad (6.45)$$

$$b = \mathbf{E} \cdot \mathbf{B} = -\frac{1}{2}\operatorname{im}(\det(F)). \quad (6.46)$$

For nonzero fields, these invariants give the standard classification:

$$b = 0 : \quad \text{electric-like when } a \text{ is positive; some frame has no magnetic field,} \quad (6.47)$$

$$b = 0 : \quad \text{magnetic-like when } a \text{ is negative; some frame has no electric field,} \quad (6.48)$$

$$b \neq 0 : \quad \text{no frame can eliminate either field.} \quad (6.49)$$

When both a and b vanish, the field is null or radiation-like: the electric and magnetic fields are orthogonal and have equal magnitude. Thus the invariants reveal physical properties that cannot be inferred from one observer's values of $\mathbf{E}$ and $\mathbf{B}$ alone. Unlike the energy density and flux below, they do not change with the observer.

6.3 Lorentz force from the field and action

The Lorentz proper-force equation is

$$m \frac{dU}{ds} = q \operatorname{re}(FU) = \frac{q}{2}(FU + UF^H). \quad (6.50)$$

The Hermitian extraction removes the imaginary magnetic-dual components. The remaining scalar and vector components are, respectively,

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot \mathbf{E} \quad (6.51)$$

$$\frac{d\mathbf{P}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (6.52)$$

The same result follows from the potential through the interacting-particle action

$$\int L dt = -m \int ds - q \int \operatorname{sc}(\Phi^*U) ds \quad (6.53)$$

$$= \int \left(-\frac{m}{\gamma} - q \operatorname{sc}(\Phi^*U/\gamma) \right) dt \quad (6.54)$$

$$= \int \left(-\frac{m}{\gamma} - q \operatorname{sc}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right) dt \quad (6.55)$$

$$= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt. \quad (6.56)$$

Thus

$$L = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}) \quad (6.57)$$

$$\partial_v L = m\gamma\mathbf{v} + q\mathbf{A} = \mathbf{P} + q\mathbf{A}. \quad (6.58)$$

The Euler–Lagrange terms are

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \frac{d\mathbf{P}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \quad (6.59)$$

$$\partial_{\mathbf{r}} L = -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}). \quad (6.60)$$

Using $(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A} = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})$ gives

$$\frac{d\mathbf{P}}{dt} = -q\partial_{\mathbf{r}} V - q\partial_t \mathbf{A} + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}) \quad (6.61)$$

$$= q(-\partial_{\mathbf{r}} V - \partial_t \mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})) \quad (6.62)$$

$$= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (6.63)$$

The Hamiltonian follows with the gauge-dependent terms cancelling:

$$H = \mathbf{v} \cdot \partial_{\mathbf{v}} L - L \quad (6.64)$$

$$= m\gamma \mathbf{v}^2 + q\mathbf{v} \cdot \mathbf{A} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \quad (6.65)$$

$$= m\gamma(\mathbf{v}^2 + \gamma^{-2}) + qV = m\gamma + qV. \quad (6.66)$$

Finally, $dH/dt = -\partial_t L$ gives

$$\frac{d(m\gamma)}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \quad (6.67)$$

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot (-\partial_{\mathbf{r}} V - \partial_t \mathbf{A}) = q\mathbf{v} \cdot \mathbf{E}. \quad (6.68)$$

To see equation (6.50) componentwise, expand

$$\frac{FU}{\gamma} = \mathbf{v} \cdot \mathbf{E} + i\mathbf{v} \cdot \mathbf{B} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + i(\mathbf{B} + \mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.69)$$

Taking the real paravector part removes the two imaginary components.

6.4 Proper acceleration from the Lorentz force

The Lorentz force is automatically compatible with the orthogonality of four-velocity and four-acceleration. Since $U^*U = 1$, cyclicity of the scalar part and $\text{sc}(F) = \text{sc}(F^{\mathbf{H}}) = 0$ give

$$\text{sc}(U^* \text{re}(FU)) = \frac{1}{2} \text{sc}(U^*FU + U^*UF^{\mathbf{H}}) = \frac{1}{2} (\text{sc}(F) + \text{sc}(F^{\mathbf{H}})) = 0. \quad (6.70)$$

Equation (6.50) therefore implies

$$\text{sc}(U^* dU/ds) = 0, \quad (6.71)$$

in agreement with the proper-acceleration development above.

The coordinate acceleration follows from the scalar and vector components of the same equation. Using $d\gamma/dt = \gamma^3 \mathbf{v} \cdot \mathbf{a}$ in the energy equation gives

$$m\gamma^3 \mathbf{v} \cdot \mathbf{a} = q\mathbf{v} \cdot \mathbf{E}. \quad (6.72)$$

Differentiating $\mathbf{P} = m\gamma\mathbf{v}$ in the momentum equation gives

$$m\gamma(\mathbf{a} + \gamma^2(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (6.73)$$

Eliminating $\mathbf{v} \cdot \mathbf{a}$ between these two equations yields

$$\mathbf{a} = \frac{q}{m\gamma}(\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v}). \quad (6.74)$$

The magnetic term is perpendicular to $\mathbf{v}$ and therefore changes momentum without contributing to the coordinate-time power $d(m\gamma)/dt = q\mathbf{v} \cdot \mathbf{E}$.

Finally, substitute this acceleration into equation (5.39). Since

$$\mathbf{v} \cdot \mathbf{a} = \frac{q}{m\gamma^3}\mathbf{v} \cdot \mathbf{E}, \quad (6.75)$$

the invariant proper-acceleration magnitude becomes

$$-\det\left(\frac{dU}{ds}\right) = \frac{q^2}{m^2}\gamma^2((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2). \quad (6.76)$$

The quantity on the right is the squared electric-field magnitude in the particle's instantaneous rest frame, multiplied by q^2/m^2 . Thus the Lorentz-force equation supplies exactly the space-like proper acceleration required by $U^*U = 1$: the electric field accelerates the particle in its rest frame, while the magnetic field enters through the frame-dependent split into electric and magnetic components.

6.5 Energy balance

Since

$$F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma} \quad (6.77)$$

$$F^H = (\mathbf{E} - \mathbf{B}i) \cdot \boldsymbol{\sigma}, \quad (6.78)$$

their ordered product is

$$FF^H = \mathbf{E}^2 + \mathbf{B}^2 + 2(\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (6.79)$$

Therefore the standard electromagnetic energy density u and Poynting vector $\mathbf{S}$ are the scalar and vector coefficients of $\frac{1}{2}FF^H$:

$$u := \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) = \frac{1}{2}\text{sc}(FF^H), \quad (6.80)$$

$$\mathbf{S} := \mathbf{E} \times \mathbf{B} = \frac{1}{2}\text{vec}(FF^H). \quad (6.81)$$

These quantities describe energy measured in the chosen frame; they are not the two Lorentz invariants contained in F^2 . The product $\frac{1}{2}FF^H$ packages energy density and energy flux into a paravector, but is not by itself the full electromagnetic stress-energy tensor. This product order gives the conventional sign of the Poynting vector directly. For comparison, reversing the order preserves the scalar coefficient and changes the sign of the vector coefficient:

$$\text{sc}(F^H F) = \text{sc}(FF^H) \quad (6.82)$$

$$\text{vec}(F^H F) = -\text{vec}(FF^H). \quad (6.83)$$

Starting only from the paravector field equation $\partial F = D^*$, its real-vector and imaginary-vector components give

$$\partial_t \mathbf{E} = \partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J} \quad (6.84)$$

$$\partial_t \mathbf{B} = -\partial_{\mathbf{r}} \times \mathbf{E}. \quad (6.85)$$

Taking their dot products with $\mathbf{E}$ and $\mathbf{B}$, respectively, yields

$$\partial_t u = \mathbf{E} \cdot \partial_t \mathbf{E} + \mathbf{B} \cdot \partial_t \mathbf{B} \quad (6.86)$$

$$= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B}) - \mathbf{E} \cdot \mathbf{J} - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \quad (6.87)$$

$$= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E}, \quad (6.88)$$

where $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$. Substitution into $\partial_t u$ therefore derives Poynting's theorem, the local electromagnetic energy-balance law:

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (6.89)$$

Here $\mathbf{J} \cdot \mathbf{E}$ is the power density transferred from the field to matter; the minus sign is the corresponding loss of field energy. Equivalently, in paravector coefficients,

$$\frac{1}{2} \partial_t \text{sc}(FF^{\mathbf{H}}) + \frac{1}{2} \partial_{\mathbf{r}} \cdot \text{vec}(FF^{\mathbf{H}}) = -\mathbf{J} \cdot \mathbf{E}. \quad (6.90)$$

7 Relativistic wave equations

7.1 Paravector inner products and expectations

For paravector amplitudes ψ, ψ' and a paravector Z , define the inner product and matrix element intrinsically by scalar extraction:

$$\langle \psi' | \psi \rangle := \text{sc}(\psi'^{\mathbf{H}} \psi), \quad (7.1)$$

$$\langle \psi' | Z | \psi \rangle := \text{sc}(\psi'^{\mathbf{H}} Z \psi). \quad (7.2)$$

The first is the $Z = 1$ case of the second, and $\langle \psi | \psi \rangle = \|\psi\|^2$. Thus normalization is $\text{sc}(\psi^{\mathbf{H}} \psi) = 1$, not $\psi^{\mathbf{H}} \psi = 1$. For an observable $Z^{\mathbf{H}} = Z$ acting from the left,

$$\langle Z \rangle_{\psi} := \frac{\langle \psi | Z | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^{\mathbf{H}} Z \psi)}{\text{sc}(\psi^{\mathbf{H}} \psi)}. \quad (7.3)$$

For normalized ψ , this reduces to $\langle Z \rangle_{\psi} = \langle \psi | Z | \psi \rangle$. Appendix A.4 shows that these intrinsic definitions become the usual trace pairing and, after minimal-ideal projection, the standard two-component column-spinor formulas.

For a paravector T with $\psi' = T\psi$,

$$\langle Z \rangle_{\psi'} = \frac{\text{sc}(\psi^{\mathbf{H}} T^{\mathbf{H}} Z T \psi)}{\text{sc}(\psi^{\mathbf{H}} T^{\mathbf{H}} T \psi)}. \quad (7.4)$$

If $T = U$, $U^{\mathbf{H}} U = 1$, and $\langle \psi | \psi \rangle = 1$, then the denominator is unity and

$$\langle Z \rangle_{\psi'} = \text{sc}(\psi^{\mathbf{H}} U^{\mathbf{H}} Z U \psi). \quad (7.5)$$

Thus a unitary rotation preserves the normalization. A finite boost is not unitary, so this positive equal-time norm is not itself Lorentz invariant; the covariant Dirac normalization instead comes from the conserved Dirac current on a spacelike hypersurface.

7.2 Physical spinors and minimal ideals

A physical two-component Pauli spinor is represented intrinsically by a minimal ideal obtained by left projection in the paravector algebra. Choose a fixed unit direction $\mathbf{u}$ and use the projectors from Section 3.6,

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (7.6)$$

They satisfy

$$\Pi_{\pm}^2 = \Pi_{\pm}, \quad (7.7)$$

$$\Pi_+ \Pi_- = 0, \quad (7.8)$$

$$\Pi_+ + \Pi_- = 1, \quad (7.9)$$

$$\Pi_{\pm}^H = \Pi_{\pm}, \quad (7.10)$$

$$\Pi_{\pm}^* = \Pi_{\mp}, \quad (7.11)$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi_{\pm} = \pm\Pi_{\pm}, \quad (7.12)$$

and hence $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$.

Project an unrestricted paravector amplitude from the left into either sector:

$$\psi'_{\pm} := \Pi_{\pm}\psi. \quad (7.13)$$

Then $\Pi_{\pm}\psi'_{\pm} = \psi'_{\pm}$ and $\Pi_{\mp}\psi'_{\pm} = 0$. Hermitian conjugation reverses the multiplied order, so the projected norm and observable matrix element are

$$\langle \psi'_{\pm} | \psi'_{\pm} \rangle = \text{sc}(\psi^H \Pi_{\pm}^H \Pi_{\pm} \psi) = \langle \psi | \Pi_{\pm} | \psi \rangle, \quad (7.14)$$

$$\langle \psi'_{\pm} | Z | \psi'_{\pm} \rangle = \text{sc}(\psi^H \Pi_{\pm}^H Z \Pi_{\pm} \psi) = \langle \psi | \Pi_{\pm} Z \Pi_{\pm} | \psi \rangle. \quad (7.15)$$

Appendix A.4 develops the faithful matrix image of this ideal and shows how it becomes one conventional two-component Pauli column. Applied to both paravector components of the Dirac block below, the same left projections select the conventional four-component Dirac sectors.

7.3 Spin carried by paravector components

Spin is determined by how an amplitude transforms under rotations, not merely by how many coefficients are used to write it. Let R be the rotation rotor defined in Section 8. The scalar, minimal-ideal, and vector carriers transform differently.

A complex scalar amplitude has no spatial direction and is unchanged by the rotor:

$$\psi'_0 = \psi_0. \quad (7.16)$$

It therefore carries spin zero. The Klein–Gordon field below is of this type; its single complex coefficient is proportional to the algebra identity.

A projected minimal-ideal amplitude contains two complex coefficients. Its associated Pauli column transforms by one-sided rotor multiplication, which is represented intrinsically by

$$\psi'_{1/2} = R\psi_{1/2}. \quad (7.17)$$

Because R contains the half-angle, a full 2π rotation gives

$$\psi'_{1/2} = -\psi_{1/2}. \quad (7.18)$$

This is the spin-one-half carrier. A Dirac amplitude consists of two such carriers coupled by the mass term; the pair still describes spin one-half, not spin zero plus spin one.

A pure vector amplitude $\mathbf{V} \cdot \boldsymbol{\sigma}$ transforms by the two-sided sandwich

$$\mathbf{V}' \cdot \boldsymbol{\sigma} = R^H(\mathbf{V} \cdot \boldsymbol{\sigma})R. \quad (7.19)$$

The two rotor signs cancel under a 2π rotation, so this three-component carrier returns to itself and carries spin one. The electromagnetic field $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ is a complex pure vector of this type. Maxwell's equations and transversality reduce a free massless electromagnetic wave to its two physical helicities; the three algebraic vector coefficients should not be confused with three independent photon polarizations.

Consequently, an unrestricted paravector $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ is reducible under rotations: its scalar component carries spin zero and its vector component carries spin one. Spin one-half does not arise by selecting some of those four paravector coefficients. It arises from the minimal-ideal projection and its one-sided transformation law.

7.4 Klein–Gordon equation

For a free particle, the mass-shell relation is $E^2 - \mathbf{P}^2 = m^2$. On a plane wave with phase $\exp(-iEt + i\mathbf{P} \cdot \mathbf{r})$, the operators $i\partial_t$ and $-i\partial_{\mathbf{r}}$ return the energy and momentum. The corresponding substitutions for a scalar amplitude ψ are therefore

$$E = i\partial_t \quad (7.20)$$

$$\mathbf{P} = -i\partial_{\mathbf{r}}. \quad (7.21)$$

Applying the mass-shell polynomial to ψ gives

$$\begin{aligned} 0 &= (E^2 - \mathbf{P}^2 - m^2)\psi \\ &= ((i\partial_t)^2 - (-i\partial_{\mathbf{r}})^2 - m^2)\psi \\ &= (-\partial_t^2 + \partial_{\mathbf{r}}^2 - m^2)\psi. \end{aligned}$$

Multiplying by -1 and defining $\square := \partial_t^2 - \partial_{\mathbf{r}}^2$ gives the Klein–Gordon equation

$$(\square + m^2)\psi = 0. \quad (7.22)$$

In an electromagnetic potential, minimal coupling replaces the free operators by

$$E = i\partial_t - qV \quad (7.23)$$

$$\mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}, \quad (7.24)$$

and the Klein–Gordon equation takes the covariant form

$$(-E^2 + \mathbf{P}^2 + m^2)\psi = 0. \quad (7.25)$$

7.5 Spin-zero low-energy limit

Denote the relativistic solution of the preceding equation by ψ' and separate its rest-energy phase by $\psi' = \exp(-imt)\psi$. The exact equation for the slowly varying amplitude becomes

$$(-E^2 - 2mE + \mathbf{P}^2)\psi = 0 \quad (7.26)$$

$$E = \frac{\mathbf{P}^2 - E^2}{2m}. \quad (7.27)$$

For a free particle the operators commute, and iteration gives

$$E = \frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + \cdots. \quad (7.28)$$

More generally, neglecting E^2 at leading nonrelativistic order in the covariant equation gives the Schrödinger equation

$$i\partial_t \psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi. \quad (7.29)$$

7.6 Dirac equation in block form

For a paravector Z , define

$$\mathbf{W}(Z) := \begin{Bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{Bmatrix}. \quad (7.30)$$

It satisfies

$$\mathbf{W}(Z)^2 = \begin{Bmatrix} Z \text{adj}(Z) & 0 \\ 0 & \text{adj}(Z) Z \end{Bmatrix} = \det(Z) \mathbf{I} \quad (7.31)$$

$$\mathbf{W}(Z)^* = \mathbf{W}(Z^*). \quad (7.32)$$

For the free field, define the Dirac operator and its mass conjugate:

$$\mathbf{D} := \mathbf{W}(i\partial) - m\mathbf{I} \quad (7.33)$$

$$\mathbf{D}' := \mathbf{W}(i\partial) + m\mathbf{I}. \quad (7.34)$$

The Dirac equation is

$$\mathbf{D}\Psi = 0. \quad (7.35)$$

Since $\mathbf{W}(i\partial)^2 = -\square \mathbf{I}$, multiplication by the mass conjugate gives

$$\mathbf{D}'\mathbf{D} = \mathbf{W}(i\partial)^2 - m^2 \mathbf{I} = -(\square + m^2) \mathbf{I}. \quad (7.36)$$

Consequently, $\mathbf{D}\Psi = 0$ implies $(\square + m^2)\Psi = 0$, so every component of the Dirac block satisfies the Klein–Gordon equation. This factorization is the direct connection between the first-order Dirac equation and the relativistic mass-shell equation. For $\Psi = \{\Psi_1, \Psi_2\}^\top$ it reads

$$i\partial\Psi_2 - m\Psi_1 = 0 \quad (7.37)$$

$$i\partial^* \Psi_1 - m\Psi_2 = 0. \quad (7.38)$$

Conversely, for $m \neq 0$, any scalar solution ψ' of $(\square + m^2)\psi' = 0$ gives a Dirac solution

$$\Psi = \left\{ \psi', \frac{i}{m} \partial^* \psi' \right\}^\top. \quad (7.39)$$

Indeed, the second component equation holds by definition, and the first reduces to $-(\square + m^2)\psi'/m = 0$. More generally, for constant complex scalars a and b , the same scalar solution gives

$$\Psi = \frac{1}{m} \begin{Bmatrix} ma + ib\partial \\ ia\partial^* + mb \end{Bmatrix} \psi'. \quad (7.40)$$

Substitution into the two component equations leaves respectively $-a(\square + m^2)\psi'/m$ and $-b(\square + m^2)\psi'/m$, so both vanish. This is a two-constant family generated from one Klein–Gordon solution, not a claim that one scalar solution parametrizes every Dirac solution. With a potential, the gauge-covariant form is

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0. \quad (7.41)$$

It is invariant under $\Phi' = \Phi + \partial^*\lambda$ and $\Psi = \exp(iq\lambda)\Psi'$.

7.7 Schrödinger–Pauli limit

Each relativistic Dirac component contains the rapid phase associated with the rest energy. For $n = 1, 2$, separate it from the slowly varying component by

$$\Psi'_n = \exp(-imt)\Psi_n. \quad (7.42)$$

For the covariant operators $E = i\partial_t - qV$ and $\mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}$, the exponential is spatially constant and therefore

$$E\Psi'_n = \exp(-imt)(E + m)\Psi_n, \quad (7.43)$$

$$\mathbf{P}\Psi'_n = \exp(-imt)\mathbf{P}\Psi_n. \quad (7.44)$$

After cancelling the common exponential, the two Dirac equations become

$$(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 = 0, \quad (7.45)$$

$$(E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 = 0. \quad (7.46)$$

Eliminating Ψ_1 gives the exact second-order equation

$$(E^2 + 2mE - \mathbf{P}^2 + q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma})\Psi_2 = 0. \quad (7.47)$$

Under the usual low-energy, slowly varying-field power counting, the E^2 term and the electric-field term contribute beyond the leading nonrelativistic approximation. Retaining the leading terms gives

$$E\Psi_2 = \frac{\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\Psi_2. \quad (7.48)$$

At this order Ψ_1 and Ψ_2 have the same leading rest-frame amplitude. Writing its faithful two-component column as Ψ and substituting the covariant operators gives the Pauli equation

$$i\partial_t\Psi = \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right)\Psi. \quad (7.49)$$

Using the spin projectors, $\Psi = \Psi_+ + \Psi_-$ with $\Psi_{\pm} = \Pi_{\pm}\Psi$. For a uniform static magnetic field, $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$, the sectors decouple:

$$i\partial_t\Psi_{\pm} = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right)\Psi_{\pm}. \quad (7.50)$$

For a varying field direction the sectors are coupled. With H denoting the Pauli Hamiltonian, direct differentiation gives

$$i\partial_t\Psi_{\pm} = i(\partial_t\Pi_{\pm})\Psi + \Pi_{\pm}H\Psi \quad (7.51)$$

$$= H\Psi_{\pm} + (i\partial_t\Pi_{\pm} + [\Pi_{\pm}, H])\Psi. \quad (7.52)$$

Because the field term commutes with its own instantaneous projectors, this is

$$\mathrm{i}\partial_t \Psi_{\pm} = H \Psi_{\pm} + S_{\pm} \quad (7.53)$$

$$S_{\pm} = \left(\mathrm{i}\partial_t \Pi_{\pm} + \frac{[\mathbf{P}^2, \Pi_{\pm}]}{2m} \right) \Psi. \quad (7.54)$$

7.8 Interacting operators and factorization

The covariant energy and momentum operators do not commute. Direct application to a test wavefunction gives

$$[E, P_m] = \mathrm{i}qE_m \quad (7.55)$$

$$[P_m, P_n] = \mathrm{i}q(\partial_m A_n - \partial_n A_m) \quad (7.56)$$

$$\mathbf{P} \times \mathbf{P} = \mathrm{i}q\mathbf{B}. \quad (7.57)$$

Therefore

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - \mathrm{i}qF, \quad (7.58)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - \mathrm{i}qF^*, \quad (7.59)$$

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = \mathrm{i}\partial^* - q\Phi, \quad (7.60)$$

$$E - \boldsymbol{\sigma} \cdot \mathbf{P} = \mathrm{i}\partial - q\Phi^*. \quad (7.61)$$

Expanding the minimally coupled Klein–Gordon operator gives the fully explicit result

$$(\square + \mathrm{i}qS - q^2 \det(\Phi) + m^2 + 2\mathrm{i}q(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})) \psi = 0. \quad (7.62)$$

The two orderings satisfy

$$0 = ((\partial + \mathrm{i}q\Phi^*)(\partial^* + \mathrm{i}q\Phi) + m^2) \Psi_1 = (-E^2 + \mathbf{P}^2 + m^2 + \mathrm{i}qF^*) \Psi_1, \quad (7.63)$$

$$0 = ((\partial^* + \mathrm{i}q\Phi)(\partial + \mathrm{i}q\Phi^*) + m^2) \Psi_2 = (-E^2 + \mathbf{P}^2 + m^2 + \mathrm{i}qF) \Psi_2. \quad (7.64)$$

These follow from the same mass-conjugate factorization used for the free Dirac equation. With the potential included, the two block operators become

$$\mathbf{D} := \mathbf{W}(\mathrm{i}\partial - q\Phi^*) - m\mathbf{I}, \quad (7.65)$$

$$\mathbf{D}' := \mathbf{W}(\mathrm{i}\partial - q\Phi^*) + m\mathbf{I}. \quad (7.66)$$

Then

$$\mathbf{D}'\mathbf{D} = - \begin{Bmatrix} -E^2 + \mathbf{P}^2 + m^2 + \mathrm{i}qF^* & 0 \\ 0 & -E^2 + \mathbf{P}^2 + m^2 + \mathrm{i}qF \end{Bmatrix}. \quad (7.67)$$

The electromagnetic field therefore prevents the interacting product from being proportional to the block identity: its two components acquire the distinct spin couplings $\mathrm{i}qF^*$ and $\mathrm{i}qF$. In the free limit $q = 0$, both couplings vanish and $-E^2 + \mathbf{P}^2 = \square$, so

$$\mathbf{D}'\mathbf{D} = -(\square + m^2)\mathbf{I}_2. \quad (7.68)$$

Thus the free Dirac equation $\mathbf{D}\Psi = 0$ recovers the Klein–Gordon equation on both components of the Dirac block.

7.9 Spin probabilities and energy sectors

The measurement projectors act from the left, so the projected amplitudes are $\psi'_\pm = \Pi_\pm \psi$ as defined above. They are orthogonal and complete:

$$\psi = \psi'_+ + \psi'_- \quad (7.69)$$

$$\langle \psi'_+ | \psi'_- \rangle = 0 \quad (7.70)$$

$$\langle \psi'_- | \psi'_+ \rangle = 0. \quad (7.71)$$

Define the real Bloch vector of a nonzero amplitude by its intrinsic expectation values,

$$s_n := \frac{\text{sc}(\psi^H \sigma_n \psi)}{\text{sc}(\psi^H \psi)} \quad (7.72)$$

$$\mathbf{s} := \{s_1, s_2, s_3\} \quad (7.73)$$

$$\|\mathbf{s}\|^2 \leq 1. \quad (7.74)$$

Equality holds for a pure two-component spin state. The probabilities for spin along and against $\mathbf{u}$ are

$$p_\pm(\mathbf{u}) := \frac{\langle \psi'_\pm | \psi'_\pm \rangle}{\langle \psi | \psi \rangle} = \frac{\text{sc}(\psi^H \Pi_\pm \Pi_\pm \psi)}{\text{sc}(\psi^H \psi)} = \frac{\text{sc}(\psi^H \Pi_\pm \psi)}{\text{sc}(\psi^H \psi)} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}). \quad (7.75)$$

Their conventional column-spinor and density-matrix forms are derived in [Appendix A.4](#).

For a state aligned with $\mathbf{u}$ and $\mathbf{u} \cdot \mathbf{u}' = \cos \alpha$, the measurement probabilities along $\mathbf{u}'$ are

$$p_+(\mathbf{u}') = \frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}') = \cos^2(\alpha/2) \quad (7.76)$$

$$p_-(\mathbf{u}') = \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}') = \sin^2(\alpha/2). \quad (7.77)$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through θ about $\mathbf{n}$, then

$$\mathbf{u} \cdot \mathbf{u}' = (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos \theta. \quad (7.78)$$

Thus $\alpha = \theta$ when $\mathbf{n} \cdot \mathbf{u} = 0$.

For fluctuations $d\mathbf{u}$ and $d\mathbf{u}'$, the squared determinant identity is

$$\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma})) = (d\mathbf{u} \cdot d\mathbf{u}')^2 + \frac{1}{4} \| [d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}] \|^2. \quad (7.79)$$

Since the first term is nonnegative,

$$\sqrt{\det((d\mathbf{u} \cdot \boldsymbol{\sigma})(d\mathbf{u}' \cdot \boldsymbol{\sigma}))} \geq \frac{1}{2} \| [d\mathbf{u} \cdot \boldsymbol{\sigma}, d\mathbf{u}' \cdot \boldsymbol{\sigma}] \|. \quad (7.80)$$

Finally, on the mass shell let

$$U = \frac{E + \mathbf{P} \cdot \boldsymbol{\sigma}}{m}. \quad (7.81)$$

Since $\det(U) = 1$ and $\mathbf{W}(U)^2 = \mathbf{I}$, the positive- and negative-energy projectors are

$$\frac{1}{2}(\mathbf{I} + \mathbf{W}(U)) \quad (7.82)$$

$$\frac{1}{2}(\mathbf{I} - \mathbf{W}(U)). \quad (7.83)$$

Each squares to itself, their product is zero, and their sum is $\mathbf{I}$. The spin projectors act with the same sign on both components of $\boldsymbol{\Psi}$, so no additional block notation is needed: their actions are

simply $\Pi_+ \Psi$ and $\Pi_- \Psi$. When $\mathbf{W}(U)$ commutes with $\Pi_\pm$, as for a compatible spin quantization axis, the four pairwise products are joint energy–spin projectors. They resolve the physical Dirac column selected in Section 7.2 as

$$\Psi = \frac{1}{2}(\mathbf{I} + \mathbf{W}(U))\Pi_+ \Psi + \frac{1}{2}(\mathbf{I} + \mathbf{W}(U))\Pi_- \Psi \quad (7.84)$$

$$+ \frac{1}{2}(\mathbf{I} - \mathbf{W}(U))\Pi_+ \Psi + \frac{1}{2}(\mathbf{I} - \mathbf{W}(U))\Pi_- \Psi. \quad (7.85)$$

Without this commutation condition the displayed sum remains a sequential decomposition, but the factors are not joint orthogonal projectors.

8 Rotations and Lorentz boosts

Throughout this section, the transformation factor T is defined from a positive generator. Thus $T = R$ for rotations and $T = L$ for boosts. The opposite action is written with T^{-1} ; it is not introduced by redefining T with a negative exponential.

8.1 Rotors

A rotation through angle θ about unit axis $\mathbf{u}$ is generated by

$$R := \exp\left(\mathbf{i} \frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} + \mathbf{u} \cdot \mathbf{i} \boldsymbol{\sigma} \sin \frac{\theta}{2} \quad (8.1)$$

$$R^{-1} = R^H = \cos \frac{\theta}{2} - \mathbf{u} \cdot \mathbf{i} \boldsymbol{\sigma} \sin \frac{\theta}{2}. \quad (8.2)$$

The sandwich action $X' = R^H X R$ fixes the scalar component and rotates the spatial vector according to Rodrigues' formula,

$$\mathbf{r}' = \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \theta)\mathbf{u}. \quad (8.3)$$

The half-angle explains the spinorial sign change $R(\theta + 2\pi) = -R(\theta)$. The corresponding column-spinor and density-matrix transformations are collected in Appendix A.4. For a spinor $\psi_R := R\psi$, expectation values transform as

$$\psi_R^H = \psi^H R^H \quad (8.4)$$

$$\langle Z \rangle_{\psi_R} = \frac{\text{sc}(\psi^H R^H Z R \psi)}{\text{sc}(\psi^H \psi)} \quad (8.5)$$

$$R^H = R^{-1}. \quad (8.6)$$

For $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$, this gives $Z' = R^H Z R = \mathbf{u}' \cdot \boldsymbol{\sigma}$.

8.2 Boosts

Let rapidity θ satisfy $\beta = \tanh \theta$ and $\gamma = \cosh \theta$. A boost along $\mathbf{u}$ is

$$L := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh \frac{\theta}{2} + \mathbf{u} \cdot \boldsymbol{\sigma} \sinh \frac{\theta}{2} \quad (8.7)$$

$$L^H = L \quad (8.8)$$

$$L^{-1} = L^* = \cosh \frac{\theta}{2} - \mathbf{u} \cdot \boldsymbol{\sigma} \sinh \frac{\theta}{2}. \quad (8.9)$$

For $\psi_L := L^{-1}\psi$, the inverse boost factor is Hermitian, so

$$\psi_L^H = \psi^H L^{-1} \quad (8.10)$$

$$\langle Z \rangle_{\psi_L} = \frac{\text{sc}(\psi^H L^{-1} Z L^{-1} \psi)}{\text{sc}(\psi^H L^{-2} \psi)}. \quad (8.11)$$

The similarity $L^{-1}ZL$ is a different operation: it preserves $\text{sc}(Z)$ and is not the spacetime or spinor-bilinear boost used here. The spacetime paravector therefore transforms by the Hermitian congruence $X' = L^{-1}XL^{-1}$, which yields the components $\mathbf{r}_{\parallel} = (\mathbf{r} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}$, $\mathbf{r}'_{\parallel} = (\mathbf{r}' \cdot \mathbf{u})\mathbf{u}$, and $\mathbf{r}'_{\perp} = \mathbf{r}' - \mathbf{r}'_{\parallel}$:

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \quad (8.12)$$

$$\mathbf{r}'_{\parallel} = \gamma(\mathbf{r}_{\parallel} - \beta t \mathbf{u}), \quad (8.13)$$

$$\mathbf{r}'_{\perp} = \mathbf{r}_{\perp}. \quad (8.14)$$

It preserves $\det(X) = t^2 - \mathbf{r}^2$. The electromagnetic field instead obeys $F' = L^{-1}FL$ for $F = (\mathbf{E} + \mathbf{B}i) \cdot \boldsymbol{\sigma}$. Define $\mathbf{E}_{\parallel} = (\mathbf{E} \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}_{\perp} = \mathbf{E} - \mathbf{E}_{\parallel}$ and likewise $\mathbf{E}'_{\parallel} = (\mathbf{E}' \cdot \mathbf{u})\mathbf{u}$, $\mathbf{E}'_{\perp} = \mathbf{E}' - \mathbf{E}'_{\parallel}$; define the four corresponding components of $\mathbf{B}$ and $\mathbf{B}'$ by the same formulas. Then

$$\mathbf{E}'_{\parallel} = \mathbf{E}_{\parallel}, \quad (8.15)$$

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \beta \mathbf{u} \times \mathbf{B}), \quad (8.16)$$

$$\mathbf{B}'_{\parallel} = \mathbf{B}_{\parallel}, \quad (8.17)$$

$$\mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - \beta \mathbf{u} \times \mathbf{E}). \quad (8.18)$$

With the same convention,

$$\partial' = L^{-1}\partial L \quad (8.19)$$

$$D'^* = L^{-1}D^*L. \quad (8.20)$$

Since L is constant and $LL^{-1} = 1$, the field equation is covariant:

$$\partial' F' = L^{-1}(\partial F)L = L^{-1}D^*L = D'^*. \quad (8.21)$$

9 Generalized spacetime and metric form

Let

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma} \quad (9.1)$$

$$dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}. \quad (9.2)$$

Writing $dX = A dt + \mathbf{B} \cdot d\mathbf{r}$ gives

$$A = \partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma} \quad (9.3)$$

$$\mathbf{B} = \partial_r f + \partial_r (\mathbf{g} \cdot \boldsymbol{\sigma}), \quad (9.4)$$

and the interval remains

$$ds^2 = dX dX^* = df^2 - d\mathbf{g}^2. \quad (9.5)$$

For a future-directed time-like curve with $df > 0$,

$$ds = \sqrt{1 - \mathbf{v}^2} df \quad (9.6)$$

$$\mathbf{v}^2 < 1. \quad (9.7)$$

Define the geometric velocity and Lorentz factor by

$$\mathbf{v} = \frac{d\mathbf{g}}{df} \quad (9.8)$$

$$\gamma = \frac{df}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}} \quad (9.9)$$

$$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (9.10)$$

With geometric acceleration $\mathbf{a} = d\mathbf{v}/ds = \gamma d^2\mathbf{g}/df^2$,

$$\frac{dU}{ds} = \gamma^3(\mathbf{v} \cdot \mathbf{a}) + (\gamma\mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (9.11)$$

For inertial motion, $dU/ds = 0$. Its scalar component gives $\mathbf{v} \cdot \mathbf{a} = 0$, and its vector component then gives $\mathbf{a} = 0$. Hence $d^2\mathbf{g}/df^2 = 0$, so $\mathbf{g} = c_1 f + c_2$ and the geometric velocity is constant. If $f' = df/dt$ and $\mathbf{g}' = d\mathbf{g}/dt$, then

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}''f' - \mathbf{g}'f''}{(f')^2} \quad (9.12)$$

$$[U, dU/ds] = 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot i\boldsymbol{\sigma}. \quad (9.13)$$

A Pauli-matrix representation

A.1 Faithful representation

Every complex element of the displayed algebra basis can be collected as

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (A.1)$$

Here $S \in \mathbb{C}$ and $\mathbf{V} \in \mathbb{C}^3$. Such an element is called a complex paravector. Its faithful 2×2 complex representation begins with

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Thus

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{Bmatrix}. \quad (A.2)$$

It preserves identity, addition, and multiplied order:

$$\text{mat}(1) = \mathbf{I}_2, \quad (A.3)$$

$$\text{mat}(Z'Z) = \text{mat}(Z')\text{mat}(Z). \quad (A.4)$$

A.2 Matrix operations and induced algebra operations

Let $\mathbf{M} = \text{mat}(Z)$ with entries $m_{jk} = a_{jk} + b_{jk}\mathbf{i}$, where $a_{jk}, b_{jk} \in \mathbb{R}$. Define transpose by

$$(\mathbf{M}^\top)_{jk} := m_{kj}. \quad (\text{A.5})$$

Define entrywise complex conjugation independently by

$$m_{jk}^{\mathbf{C}} := a_{jk} - b_{jk}\mathbf{i} \quad (\text{A.6})$$

$$\mathbf{M}^{\mathbf{C}} := \begin{Bmatrix} m_{11}^{\mathbf{C}} & m_{12}^{\mathbf{C}} \\ m_{21}^{\mathbf{C}} & m_{22}^{\mathbf{C}} \end{Bmatrix}. \quad (\text{A.7})$$

The Hermitian transpose is then

$$\mathbf{M}^{\mathbf{H}} := (\mathbf{M}^{\mathbf{C}})^\top = (\mathbf{M}^\top)^{\mathbf{C}}. \quad (\text{A.8})$$

Define the algebra operation $^{\mathbf{H}}$ through the faithful representation by

$$\text{mat}(Z^{\mathbf{H}}) := \mathbf{M}^{\mathbf{H}}. \quad (\text{A.9})$$

For $n = 1, 2, 3$, its action on every basis type is

$$\text{mat}(1)^{\mathbf{H}} = \mathbf{I}_2, \quad (\text{A.10})$$

$$(\mathbf{i}\mathbf{I}_2)^{\mathbf{H}} = -\mathbf{i}\mathbf{I}_2, \quad (\text{A.11})$$

$$\text{mat}(\sigma_n)^{\mathbf{H}} = \text{mat}(\sigma_n), \quad (\text{A.12})$$

$$\text{mat}(\mathbf{i}\sigma_n)^{\mathbf{H}} = -\text{mat}(\mathbf{i}\sigma_n). \quad (\text{A.13})$$

Hence $1^{\mathbf{H}} = 1$, $\mathbf{i}^{\mathbf{H}} = -\mathbf{i}$, $\sigma_n^{\mathbf{H}} = \sigma_n$, and $(\mathbf{i}\sigma_n)^{\mathbf{H}} = -\mathbf{i}\sigma_n$. Hermitian conjugation changes matrix-product order. Therefore faithfulness gives

$$\text{mat}((\sigma_n\sigma_m)^{\mathbf{H}}) = (\text{mat}(\sigma_n)\text{mat}(\sigma_m))^{\mathbf{H}} \quad (\text{A.14})$$

$$= \text{mat}(\sigma_m)^{\mathbf{H}}\text{mat}(\sigma_n)^{\mathbf{H}} = \text{mat}(\sigma_m\sigma_n), \quad (\text{A.15})$$

and hence

$$(\sigma_n\sigma_m)^{\mathbf{H}} = \sigma_m^{\mathbf{H}}\sigma_n^{\mathbf{H}} = \sigma_m\sigma_n. \quad (\text{A.16})$$

Thus $^{\mathbf{H}}$ is precisely the Hermitian operator defined in Section 2. To define the distinct sigma conjugation, let $\mathbf{S} := \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}$ and $\mathbf{S}^2 = -\mathbf{I}_2$, so that $\mathbf{S}^{-1} = -\mathbf{S}$, and set

$$\mathbf{M}^* := \mathbf{S}^{-1}\mathbf{M}^{\mathbf{C}}\mathbf{S}, \quad (\text{A.17})$$

$$\text{mat}(Z^*) := \mathbf{M}^*. \quad (\text{A.18})$$

For $n = 1, 2, 3$, this gives the required action on every basis type:

$$\text{mat}(1)^* = \mathbf{S}^{-1}\mathbf{I}_2^{\mathbf{C}}\mathbf{S} = \mathbf{I}_2, \quad (\text{A.19})$$

$$(\mathbf{i}\mathbf{I}_2)^* = \mathbf{S}^{-1}(\mathbf{i}\mathbf{I}_2)^{\mathbf{C}}\mathbf{S} = -\mathbf{i}\mathbf{I}_2, \quad (\text{A.20})$$

$$\text{mat}(\sigma_n)^* = \mathbf{S}^{-1}\text{mat}(\sigma_n)^{\mathbf{C}}\mathbf{S} = -\text{mat}(\sigma_n), \quad (\text{A.21})$$

$$\text{mat}(\mathbf{i}\sigma_n)^* = \mathbf{S}^{-1}\text{mat}(\mathbf{i}\sigma_n)^{\mathbf{C}}\mathbf{S} = \text{mat}(\mathbf{i}\sigma_n). \quad (\text{A.22})$$

The operation also preserves multiplied order. Since entrywise conjugation preserves order,

$$(MN)^* = S^{-1} M^C N^C S \quad (\text{A.23})$$

$$= (S^{-1} M^C S)(S^{-1} N^C S) = M^* N^*. \quad (\text{A.24})$$

Therefore

$$(MN)^* = M^* N^*. \quad (\text{A.25})$$

The four operations must remain distinct:

symbol	matrix operation	relation to the algebra
M^T	transpose	changes matrix-product order
M^C	entrywise complex conjugation	sends i to $-i$
M^H	Hermitian transpose	$\text{mat}(Z^H)$
M^*	sigma conjugation	$\text{mat}(Z^*)$

Their product-order rules are

$$(MN)^T = N^T M^T, \quad (\text{A.26})$$

$$(MN)^C = M^C N^C, \quad (\text{A.27})$$

$$(MN)^H = N^H M^H. \quad (\text{A.28})$$

Sigma conjugation preserves order by equation (A.25). Thus the induced algebra operations satisfy

$$(Z^H)^H = Z, \quad (\text{A.29})$$

$$(ZZ')^H = Z'^H Z^H, \quad (\text{A.30})$$

$$(Z^*)^* = Z, \quad (\text{A.31})$$

$$(ZZ')^* = Z^* Z'^*, \quad (\text{A.32})$$

$$Z^{*H} = Z^{H*}. \quad (\text{A.33})$$

A.2.1 Trace, determinant, and adjugate

For $M = \text{mat}(Z)$, define the matrix trace and determinant by

$$\text{tr}(M) := m_{11} + m_{22} \quad (\text{A.34})$$

$$\det(M) := m_{11}m_{22} - m_{12}m_{21}. \quad (\text{A.35})$$

The corresponding paravector operations are

$$\text{tr}(Z) := \text{tr}(M) = 2S, \quad (\text{A.36})$$

$$\det(Z) := \det(M). \quad (\text{A.37})$$

Faithfulness transfers the matrix identities to the algebra:

$$\text{tr}(Z + Z') = \text{tr}(Z) + \text{tr}(Z'), \quad (\text{A.38})$$

$$\text{tr}(cZ) = c \text{tr}(Z), \quad (\text{A.39})$$

$$\text{tr}(ZZ') = \text{tr}(Z'Z), \quad (\text{A.40})$$

$$\det(ZZ') = \det(Z) \det(Z'), \quad (\text{A.41})$$

$$\det(cZ) = c^2 \det(Z). \quad (\text{A.42})$$

The last identity holds for $c \in \mathbb{C}$. For a 2×2 matrix and its paravector preimage, define

$$\text{adj}(\mathbf{M}) := \text{tr}(\mathbf{M})\mathbf{I}_2 - \mathbf{M}, \quad (\text{A.43})$$

$$\text{mat}(\text{adj}(Z)) := \text{adj}(\mathbf{M}). \quad (\text{A.44})$$

Direct comparison with the induced algebra operators gives

$$\text{adj}(Z) = Z^{*\text{H}} \quad (\text{A.45})$$

$$\text{mat}(\text{adj}(Z)) = \text{adj}(\mathbf{M}) = (\mathbf{M}^*)^{\text{H}}. \quad (\text{A.46})$$

Equivalently,

$$\text{adj}(Z) = \text{tr}(Z) - Z \quad (\text{A.47})$$

$$\text{tr}(Z) = Z + \text{adj}(Z) = 2S. \quad (\text{A.48})$$

The adjugate changes product order and obeys

$$\text{adj}(ZZ') = \text{adj}(Z')\text{adj}(Z), \quad (\text{A.49})$$

$$\text{adj}(\text{adj}(Z)) = Z, \quad (\text{A.50})$$

$$\text{adj}(Z)Z = Z\text{adj}(Z) = \det(Z), \quad (\text{A.51})$$

$$\text{mat}(\text{adj}(Z)Z) = \det(\mathbf{M})\mathbf{I}_2. \quad (\text{A.52})$$

Consequently, when $\det(Z) \neq 0$,

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)}. \quad (\text{A.53})$$

Finally, Cayley–Hamilton gives

$$\mathbf{M}^2 - \text{tr}(\mathbf{M})\mathbf{M} + \det(\mathbf{M})\mathbf{I}_2 = 0, \quad (\text{A.54})$$

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \quad (\text{A.55})$$

A.3 Algebra–matrix crosswalk

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, define

$$\text{sc}(Z) := S \quad (\text{A.56})$$

$$\text{vec}(Z) := \mathbf{V}. \quad (\text{A.57})$$

Thus $\text{vec}(Z)$ is the coefficient vector, while its algebra element is $\mathbf{V} \cdot \boldsymbol{\sigma}$.

For $\mathbf{M} = \text{mat}(Z)$, the extraction formulas are

$$\text{sc}(Z) = S = \frac{1}{2} \text{tr}(\mathbf{M}), \quad (\text{A.58})$$

$$\text{mat}(\text{sc}(Z)) = S\mathbf{I}_2, \quad (\text{A.59})$$

$$(\text{vec}(Z))_n = V_n = \frac{1}{2} \text{tr}(\boldsymbol{\sigma}_n \mathbf{M}), \quad (\text{A.60})$$

$$\text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) = \mathbf{M} - \frac{1}{2} \text{tr}(\mathbf{M})\mathbf{I}_2. \quad (\text{A.61})$$

Define real and imaginary paravector extraction through Hermitian splitting:

$$\text{mat}(\text{re}(Z)) := \frac{1}{2}(\mathbf{M} + \mathbf{M}^{\text{H}}), \quad (\text{A.62})$$

$$\text{mat}(\text{im}(Z)) := \frac{\mathbf{M} - \mathbf{M}^{\text{H}}}{2i}. \quad (\text{A.63})$$

By construction, the algebra and matrix operators agree through

$$\text{mat}(Z^{\text{H}}) = \mathbf{M}^{\text{H}} \quad (\text{A.64})$$

$$\text{mat}(Z^*) = \mathbf{M}^* \quad (\text{A.65})$$

$$\text{mat}(Z^{*\text{H}}) = (\mathbf{M}^*)^{\text{H}}. \quad (\text{A.66})$$

The matrix-only operations $^{\text{C}}$ and $^{\text{T}}$ are related to the algebra operations by

$$\mathbf{M}^{\text{C}} = \mathbf{S} \mathbf{M}^* \mathbf{S}^{-1} \quad (\text{A.67})$$

$$\mathbf{M}^{\text{T}} = \mathbf{S} \text{mat}(Z^{*\text{H}}) \mathbf{S}^{-1}. \quad (\text{A.68})$$

Finally, scalar extraction of a product and the paravector norm are

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}_{Z'} \mathbf{M}_Z), \quad (\text{A.69})$$

$$\|Z\|^2 := \frac{1}{2} \text{tr}(\mathbf{M}^{\text{H}} \mathbf{M}) = \frac{1}{2} \text{tr}(\mathbf{M} \mathbf{M}^{\text{H}}). \quad (\text{A.70})$$

A.4 Matrix amplitudes and column spinors

This subsection translates the intrinsic paravector spinor development of Section 7.2 into the familiar matrix and column languages. For a paravector amplitude ψ , write

$$\mathbf{M}_\psi := \text{mat}(\psi). \quad (\text{A.71})$$

Faithfulness and Hermitian compatibility give

$$\text{mat}(\psi'^{\text{H}} Z \psi) = \mathbf{M}_{\psi'}^{\text{H}} \text{mat}(Z) \mathbf{M}_\psi. \quad (\text{A.72})$$

Scalar extraction is half the matrix trace, so the intrinsic definitions (7.1) and (7.2) become

$$\langle \psi' | \psi \rangle = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^{\text{H}} \mathbf{M}_\psi), \quad (\text{A.73})$$

$$\langle \psi' | Z | \psi \rangle = \frac{1}{2} \text{tr}(\mathbf{M}_{\psi'}^{\text{H}} \text{mat}(Z) \mathbf{M}_\psi). \quad (\text{A.74})$$

Thus $\text{mat}(\psi^{\text{H}} \psi) = \mathbf{M}_\psi^{\text{H}} \mathbf{M}_\psi$ is generally Hermitian rather than scalar; the scalar inner product is obtained by taking half its trace.

For the fixed unit direction $\mathbf{u}$, the paravector projectors have matrix images

$$\text{mat}(\Pi_\pm) = \frac{1}{2} (\mathbf{I}_2 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (\text{A.75})$$

Let $\mathbf{v}_+$ be a normalized positive eigenvector,

$$(\mathbf{u} \cdot \boldsymbol{\sigma}) \mathbf{v}_+ = \mathbf{v}_+ \quad (\text{A.76})$$

$$\mathbf{v}_+^{\text{H}} \mathbf{v}_+ = 1. \quad (\text{A.77})$$

A convenient phase choice, valid for $u_3 \neq -1$, is

$$\mathbf{v}_+ = \frac{\{1 + u_3, u_1 + \text{i}u_2\}^{\text{T}}}{\sqrt{2(1 + u_3)}}. \quad (\text{A.78})$$

At the excluded direction $\mathbf{u} = -\mathbf{e}_3$, use

$$\mathbf{v}_+ = \{0, 1\}^\top. \quad (\text{A.79})$$

Then

$$\text{mat}(\Pi_+) = \mathbf{v}_+ \mathbf{v}_+^\text{H}. \quad (\text{A.80})$$

For the projected amplitude $\psi' = \Pi_+ \psi$, define

$$\mathbf{M}_{\psi'} := \text{mat}(\psi') = \text{mat}(\Pi_+) \mathbf{M}_\psi \quad (\text{A.81})$$

$$\boldsymbol{\chi} := \mathbf{M}_{\psi'}^\text{H} \mathbf{v}_+. \quad (\text{A.82})$$

The algebra, rank-one matrix, and two-component column are related by

$$\boldsymbol{\chi} = \mathbf{M}_{\psi'}^\text{H} \mathbf{v}_+ \quad (\text{A.83})$$

$$\mathbf{M}_{\psi'} = \mathbf{v}_+ \boldsymbol{\chi}^\text{H}. \quad (\text{A.84})$$

Thus ψ' and $\boldsymbol{\chi} \in \mathbb{C}^2$ contain the same two complex components. The matrix $\mathbf{M}_{\psi'}$ has rank one and has $\boldsymbol{\chi}^\text{H}$ as its nonzero row precisely when the basis begins with $\mathbf{v}_+$. Conversely, $\mathbf{v}_+ \boldsymbol{\chi}^\text{H}$ uniquely determines ψ' . Moreover,

$$\boldsymbol{\chi}^\text{H} \boldsymbol{\chi} = \text{tr}(\mathbf{M}_{\psi'}^\text{H} \mathbf{M}_{\psi'}) = 2 \text{sc}(\psi'^\text{H} \psi'), \quad (\text{A.85})$$

For an observable acting from the left, the intrinsic matrix element is

$$2 \text{sc}(\psi'^\text{H} Z \psi') = \text{tr}(\mathbf{M}_{\psi'}^\text{H} \text{mat}(Z) \mathbf{M}_{\psi'}) = (\boldsymbol{\chi}^\text{H} \boldsymbol{\chi})(\mathbf{v}_+^\text{H} \text{mat}(Z) \mathbf{v}_+). \quad (\text{A.86})$$

The two left-projected matrix amplitudes are

$$\mathbf{M}_{\psi', \pm} := \text{mat}(\Pi_\pm) \mathbf{M}_\psi. \quad (\text{A.87})$$

Consequently, the intrinsic probabilities in (7.75) have the matrix form

$$p_\pm(\mathbf{u}) = \frac{\text{tr}(\mathbf{M}_\psi^\text{H} \text{mat}(\Pi_\pm)^\text{H} \text{mat}(\Pi_\pm) \mathbf{M}_\psi)}{\text{tr}(\mathbf{M}_\psi^\text{H} \mathbf{M}_\psi)} = \frac{\text{tr}(\mathbf{M}_\psi^\text{H} \text{mat}(\Pi_\pm) \mathbf{M}_\psi)}{\text{tr}(\mathbf{M}_\psi^\text{H} \mathbf{M}_\psi)}. \quad (\text{A.88})$$

A.5 Quaternion subalgebra

The real span of the scalar and bivector basis elements is a quaternion subalgebra of the complex paravectors. Define its three imaginary units by

$$q_1 = -\sigma_2 \sigma_3 = -\mathbf{i} \sigma_1 \quad (\text{A.89})$$

$$q_2 = -\sigma_3 \sigma_1 = -\mathbf{i} \sigma_2 \quad (\text{A.90})$$

$$q_3 = -\sigma_1 \sigma_2 = -\mathbf{i} \sigma_3, \quad (\text{A.91})$$

or, collectively, $\mathbf{q} = -\mathbf{i} \boldsymbol{\sigma}$. For $n, m \in \{1, 2, 3\}$, the sigma multiplication rule gives

$$q_n q_m = -\delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} q_k, \quad (\text{A.92})$$

Consequently,

$$q_1^2 = q_2^2 = q_3^2 = -1 \quad (\text{A.93})$$

$$q_1 q_2 = q_3 \quad (\text{A.94})$$

$$q_1 q_2 q_3 = -1, \quad (\text{A.95})$$

with the remaining products fixed by anticommutation. For $a \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^3$, define the real paravector

$$Q = a + \mathbf{A} \cdot \mathbf{q} = a - \mathbf{iA} \cdot \boldsymbol{\sigma}. \quad (\text{A.96})$$

These are precisely the quaternions embedded in the paravector algebra. For a second quaternion $Q' = a' + \mathbf{A}' \cdot \mathbf{q}$, equation (A.92) gives the familiar product

$$QQ' = (aa' - \mathbf{A} \cdot \mathbf{A}') + (a\mathbf{A}' + a'\mathbf{A} + \mathbf{A} \times \mathbf{A}') \cdot \mathbf{q}. \quad (\text{A.97})$$

Hermitian conjugation reverses every bivector, while sigma conjugation fixes it:

$$q_n^{\text{H}} = -q_n \quad (\text{A.98})$$

$$q_n^* = q_n. \quad (\text{A.99})$$

Quaternion conjugation is therefore already the Hermitian operation,

$$Q^{\text{H}} = a - \mathbf{A} \cdot \mathbf{q} \quad (\text{A.100})$$

$$Q^* = Q \quad (\text{A.101})$$

$$\text{adj}(Q) = Q^{\text{H}}. \quad (\text{A.102})$$

It follows that

$$QQ^{\text{H}} = Q^{\text{H}}Q = a^2 + \mathbf{A}^2 = \det(Q) \quad (\text{A.103})$$

$$Q^{-1} = \frac{Q^{\text{H}}}{a^2 + \mathbf{A}^2} \quad (\text{A.104})$$

for nonzero Q .

The faithful matrix representation sends the quaternion units to

$$\text{mat}(q_1) = \begin{Bmatrix} 0 & -\mathbf{i} \\ -\mathbf{i} & 0 \end{Bmatrix} \quad (\text{A.105})$$

$$\text{mat}(q_2) = \begin{Bmatrix} 0 & -1 \\ 1 & 0 \end{Bmatrix} \quad (\text{A.106})$$

$$\text{mat}(q_3) = \begin{Bmatrix} -\mathbf{i} & 0 \\ 0 & \mathbf{i} \end{Bmatrix}. \quad (\text{A.107})$$

Hence equation (A.96) becomes

$$\text{mat}(Q) = a\mathbf{I}_2 - \mathbf{iA} \cdot \boldsymbol{\sigma} = \begin{Bmatrix} a - \mathbf{iA}_3 & -A_2 - \mathbf{iA}_1 \\ A_2 - \mathbf{iA}_1 & a + \mathbf{iA}_3 \end{Bmatrix}. \quad (\text{A.108})$$

The algebraic conjugation and norm become the usual matrix identities

$$\text{mat}(Q^{\text{H}}) = \text{mat}(Q)^{\text{H}}, \quad (\text{A.109})$$

$$\text{mat}(Q)^{\text{H}} \text{mat}(Q) = \text{mat}(Q) \text{mat}(Q)^{\text{H}} = (a^2 + \mathbf{A}^2)\mathbf{I}_2, \quad (\text{A.110})$$

$$\det(\text{mat}(Q)) = a^2 + \mathbf{A}^2. \quad (\text{A.111})$$

Thus a unit quaternion is represented by a unitary matrix of determinant one.

Finally, for a unit axis $\mathbf{u}$, let

$$Q = \cos \frac{\theta}{2} + \mathbf{u} \cdot \mathbf{q} \sin \frac{\theta}{2}. \quad (\text{A.112})$$

Since $\mathbf{q} = -\mathbf{i}\boldsymbol{\sigma}$, this quaternion is R^{H} for the rotor R defined in Section 8. The quaternion sandwich

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^{\text{H}} \quad (\text{A.113})$$

therefore reproduces the same Rodrigues rotation.

B Reference summary

This index collects the principal definitions and results in natural units, with signature $(+, -, -, -)$. Topic titles and equation numbers link to their full statements and derivations.

Algebra and vector operations.

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9),$$

$$[\sigma_n, \sigma_m] = 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k \quad (1.9),$$

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma},$$

$$\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A},$$

$$\|\mathbf{Z}\|^2 = \|\mathbf{S}\|^2 + \|\mathbf{V}\|^2,$$

$$\|\mathbf{V}\|^2 = \mathbf{V} \cdot \mathbf{V}^* = \mathbf{V}^* \cdot \mathbf{V} \quad (3.86).$$

Hermitian and sigma-conjugate operators.

$$(ZZ')^* = Z^* Z'^*,$$

$$(ZZ')^H = Z'^H Z^H,$$

$$\text{adj}(Z) = Z^{*H},$$

$$\text{adj}(ZZ') = Z'^{*H} Z^{*H} \quad (3.13).$$

Paravectors, extraction, and products. For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = (a + bi) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}$,

$$S = \text{sc}(Z) = \frac{1}{2}(Z + Z^{*H}),$$

$$\mathbf{V} \cdot \boldsymbol{\sigma} = \frac{1}{2}(Z - Z^{*H}),$$

$$a = \text{sc}(\text{re}(Z)),$$

$$b = \text{sc}(\text{im}(Z)),$$

$$\mathbf{A} = \text{vec}(\text{re}(Z)),$$

$$\mathbf{B} = \text{vec}(\text{im}(Z)),$$

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma},$$

$$\det(Z) = \text{adj}(Z) Z = Z \text{adj}(Z) = S^2 - \mathbf{V}^2,$$

$$Z^{-1} = \frac{\text{adj}(Z)}{\det(Z)},$$

$$Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0,$$

$$\lambda_{\pm} = S \pm \sqrt{\mathbf{V}^2}.$$

See the product (3.69), determinant (3.15), and quadratic identity (3.56).

Projection operators and functions.

$$\begin{aligned}
\Pi_{\pm} &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
\Pi_{\pm}^2 &= \Pi_{\pm}, \\
\Pi_+ \Pi_- &= 0, \\
f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_-, \\
X &= \lambda_+ \Pi_+ + \lambda_- \Pi_-, \\
f(X) &= f(\lambda_+)\Pi_+ + f(\lambda_-)\Pi_-.
\end{aligned}$$

See the general complex-paravector rule (3.155) and ladder operators in Section 3.6.1.

Space–time kinematics and action.

$$\begin{aligned}
dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\
ds^2 &= dX dX^* = dt^2 - d\mathbf{r}^2 & (5.3), \\
\gamma &= (1 - \mathbf{v}^2)^{-1/2}, \\
ds &= dt/\gamma, \\
\frac{d}{ds} &= \gamma \frac{d}{dt}, \\
U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\
mU &= E + \mathbf{P} \cdot \boldsymbol{\sigma}, \\
E^2 - \mathbf{P}^2 &= m^2 & (5.24), \\
\int L dt &= -m \int ds, \\
L &= -m/\gamma, \\
H &= m\gamma, \\
K &= H - m = m(\gamma - 1) & (5.66).
\end{aligned}$$

Electromagnetism.

$$\begin{aligned}
\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\
F &= (\mathbf{E} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, \\
D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma}, \\
\partial F &= D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma} \tag{6.4}, \\
\Box &= \partial^* \partial, \\
\partial \Phi &= S + F^*, \\
\mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\
\mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}, \\
\rho &= \Box V - \partial_t S \tag{6.29}, \\
\mathbf{J} &= \Box \mathbf{A} + \partial_{\mathbf{r}} S, \\
\frac{1}{2} F^2 &= \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) + \mathbf{i} \mathbf{E} \cdot \mathbf{B}, \\
\frac{1}{2} F F^{\mathbf{H}} &= \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
m \frac{dU}{ds} &= q \operatorname{re}(FU), \\
\frac{d\mathbf{P}}{dt} &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{6.50}, \\
\mathbf{a} &= \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v}) \tag{6.74}.
\end{aligned}$$

See the field invariants (6.46) and Poynting balance (6.89). The connection to invariant proper acceleration is developed in Section 6.4.

Wave equations, spin, and energy sectors.

$$\begin{aligned}
\langle \psi' \mid \psi \rangle &= \operatorname{sc}(\psi'^{\mathbf{H}} \psi), \\
\langle \psi' \mid Z \mid \psi \rangle &= \operatorname{sc}(\psi'^{\mathbf{H}} Z \psi), \\
\langle Z \rangle_{\psi} &= \frac{\langle \psi \mid Z \mid \psi \rangle}{\langle \psi \mid \psi \rangle} = \frac{\operatorname{sc}(\psi^{\mathbf{H}} Z \psi)}{\operatorname{sc}(\psi^{\mathbf{H}} \psi)}, \\
(\Box + m^2)\psi &= 0, \\
\mathbf{W}(Z) &= \begin{Bmatrix} 0 & Z \\ \operatorname{adj}(Z) & 0 \end{Bmatrix}, \\
(\mathbf{W}(\mathbf{i}\partial) - m\mathbf{I})\Psi &= 0, \\
(\mathbf{W}(\mathbf{i}\partial - q\Phi^*) - m\mathbf{I})\Psi &= 0, \\
\mathbf{i}\partial_t \Psi &= \left(\frac{(-\mathbf{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m} \mathbf{B} \cdot \boldsymbol{\sigma} \right) \Psi, \\
\Pi_{\pm} &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\
p_{\pm}(\mathbf{u}) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{s}), \\
\text{positive-energy projector:} & \quad \frac{1}{2}(\mathbf{I} + \mathbf{W}(U)), \\
\text{negative-energy projector:} & \quad \frac{1}{2}(\mathbf{I} - \mathbf{W}(U)), \\
\text{positive-spin projection:} & \quad \Pi_+ \Psi, \\
\text{negative-spin projection:} & \quad \Pi_- \Psi.
\end{aligned}$$

See the paravector inner product (7.1), minimal-ideal development in Section 7.2, spin-carrier comparison in Section 7.3, Klein–Gordon equation (7.22), Dirac equations (7.35) and (7.41), general Klein–Gordon lift (7.40), Pauli equation (7.49), and spin probability (7.75).

Rotations and boosts.

$$\begin{aligned}
R &= \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \\
R^H &= R^{-1}, \\
Z' &= R^H Z R, \\
L &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \\
L^H &= L, \\
X' &= L^{-1} X L^{-1}, \\
\langle Z \rangle_{\psi_T} &= \frac{\text{sc}(\psi^H T^H Z T \psi)}{\text{sc}(\psi^H T^H T \psi)}.
\end{aligned}$$

See the bilinear rule (7.4) and varying-frame derivatives in Section 5.2.

Matrix representation in Appendix A.

$$\text{mat}(Z) = \begin{Bmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{Bmatrix} \quad (\text{A.2}),$$

$$\text{mat}(ZZ') = \text{mat}(Z) \text{mat}(Z') \quad (\text{A.4}),$$

$$\text{mat}(Z^H) = \text{mat}(Z)^H,$$

$$\text{mat}(Z^*) = \mathbf{S}^{-1} \text{mat}(Z)^c \mathbf{S} \quad (\text{A.18}),$$

$$\text{mat}(\text{adj}(Z)) = \text{adj}(\text{mat}(Z)),$$

$$\det(Z) = \det(\text{mat}(Z)) \quad (\text{A.52}).$$

The minimal-ideal column map is equation (A.83).

Quaternion subalgebra in Appendix A.

$$\begin{aligned}
\mathbf{q} &= -i\boldsymbol{\sigma}, \\
Q^H &= a - \mathbf{A} \cdot \mathbf{q}, \\
QQ^H &= a^2 + \mathbf{A}^2.
\end{aligned}$$

See the quaternion product (A.92), its paravector embedding (A.96), and its faithful matrix form (A.108).

Generalized spacetime and metric form.

$$\begin{aligned}
X &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, \\
ds^2 &= df^2 - d\mathbf{g}^2, \\
\mathbf{v} &= d\mathbf{g}/df, \\
U &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).
\end{aligned}$$

C Conventions and cautions

- The same symbol $i = \sigma_1\sigma_2\sigma_3$ is both the algebra’s pseudoscalar and, in the Pauli representation, the ordinary complex unit.
- Sigma conjugation $(\cdot)^*$, the Hermitian operator $(\cdot)^H$, and their composition are distinct. In the Pauli representation, the Hermitian operator corresponds to Hermitian adjunction.
- $\det(Z)$ is an indefinite quadratic form and may be complex for complex Z .
- Differential operators act on everything to their right unless parentheses state otherwise. This matters in products involving fields.
- Vectors and matrices are bold, and only vectors and matrices are bold. In particular, the electromagnetic fields formerly written as calligraphic or italic E and B are the upright bold vectors $\mathbf{E}$ and $\mathbf{B}$, while scalar energy remains the unbold symbol E . Blackboard bold is retained only for the number sets $\mathbb{R}$ and $\mathbb{C}$.
- The action is written as $\int L dt$, avoiding conflict with the scalar part S . The Lagrangian L and Hamiltonian H use the regular math font. The Dirac block function is the upright bold $\mathbf{W}(\cdot)$, and a vector square is explicitly defined by $\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}$.
- Curly braces delimit explicit vector-entry lists and matrices, while commutators alone use square brackets. Parentheses delimit functions, algebraic grouping, and order-sensitive operator products. Redundant grouping parentheses were removed.
- The extraction functions $\text{sc}(\cdot)$, $\text{vec}(\cdot)$, $\text{re}(\cdot)$, and $\text{im}(\cdot)$ replace `scalar`, `vector`, and their capitalized variants. They are defined before use. The unused expectation abbreviation `Exp` was removed.
- The original metric notation $\langle\langle Z \rangle\rangle^2$ and determinant bars were replaced by $\det(Z)$. Double bars are reserved for squared norms, with

$$\|Z\|^2 = \|S\|^2 + \|\mathbf{V}\|^2.$$

Absolute-value bars are not used for paravector norms. Entrywise complex-conjugation notation is confined to the matrix appendix.

- Sigma conjugation retains the splat $*$. The operation originally written R , and elsewhere as a dagger, is written H because it reverses multiplied sigma order and becomes Hermitian adjunction in the matrix representation. The abbreviation $\text{adj}(Z) := Z^{*H}$ is defined early and used where it shortens formulas.
- The matrix appendix uses matching upright sans-serif capitals T , C , and H for transpose, entrywise complex conjugation, and Hermitian transpose. It distinguishes these operations from sigma conjugation and states which operations reverse product order.
- Plus/minus labels are subscripts $(\pm, +, -)$, not superscripts or parenthesized labels. Parallel and perpendicular components use $\parallel$ and $\perp$ rather than `par` and `prp`; their projection formulas are restated where needed.
- Spatial derivatives use ∂_r rather than ∇ , and no overbars are used. Literal or malformed source commands such as `quad` and `frac` were converted to formatted $\text{\LaTeX}$.
- Long product expansions were consolidated while retaining their distinct components and results. Expressions otherwise preserve the compact ordering of the source, for example $\text{sc}(\Phi^*U)$, without an extra pair of grouping parentheses.